

Acute Activation of Genes Through Transcriptional Condensates Impact Non-target Genes in a Chromatin Domain

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eLife Assessment

The authors use single molecule imaging and in vivo loop-capture genomic approaches to investigate estrogen mediated enhancer-target gene activation in human cancer cells. These potentially **important** results suggest that ER-alpha can, in a temporal delay, activate a non-target gene TFF3, which is in proximity to the main target gene TFF1, even though the estrogen responsive enhancer does not loop with the TFF3 promoter. To explain these results, the authors invoke a transcriptional condensate model. The reviewers were split on the strength and interpretation of the evidence presented, which is considered **incomplete** at this stage. We encourage a revision which buttresses the findings with additional control experiments and careful consideration of alternative explanations and mathematical models. Further, the depth of the discussion on existing literature could be improved. This work will be of interest to those studying transcriptional gene regulation and hormone-aggravated cancers.

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Abstract

Transcription activation of genes by estrogens is driven by enhancers, which are often located within the same Topologically Associating Domain (TAD) as non-targeted promoters. We investigated how acute enhancer-driven activation affects neighbouring non-target genes within the same TAD. Using single-molecule RNA FISH (smFISH), we tracked the transcription of *TFF1* (enhancer-targeted) and *TFF3* (non-targeted) during estrogen stimulation. We observed mutually exclusive expression patterns: *TFF1* expression peaked at 1 hour, while *TFF3* reached its peak at 3 hours, after *TFF1*'s activation had diminished. Chromatin looping data indicated that the enhancer loops with *TFF1* but not *TFF3*, suggesting that *TFF3* upregulation is not due to direct enhancer-promoter interactions. CRISPR deletion of the *TFF1* enhancer and 1,6-hexanediol (HD) exposure revealed that the *TFF1* enhancer:promoter undergo Liquid-Liquid Phase Separation (LLPS), which sequesters the transcriptional

machinery and inhibits *TFF3* expression. As estrogen signalling wanes or LLPS is disrupted, *TFF1* expression declines while *TFF3* expression increases. Our findings reveal that enhancer-driven activation can indirectly influence neighbouring genes, highlighting a dynamic shift in gene expression as signalling progresses.

Introduction

Acute transcriptional activation of ligand-induced genes drive downstream signalling response. Similar to development-specific genes, signalling induced genes are also driven by enhancers (Hah et al., 2013 [↗](#); Li et al., 2013 [↗](#); Liu et al., 2014 [↗](#); Uyehara & Apostolou, 2023 [↗](#)). Upon binding with ligand induced transcription factors (TFs), these enhancers loop with their target promoters mostly, in the same TAD (Buecker et al., 2014 [↗](#); Bulger & Groudine, 1999 [↗](#); Chepelev et al., 2012 [↗](#); Furlong & Levine, 2018 [↗](#); Oh et al., 2021 [↗](#); Panigrahi & O'Malley, 2021 [↗](#); Ptashne, 1986 [↗](#); Sanyal et al., 2012 [↗](#); Yan et al., 2018 [↗](#)). Enhancer: promoter pairing is thought to be specific and forms the basis of noise-free gene activation of a subset of genes crucial for signalling response (Bojcsuk et al., 2017 [↗](#); Chen et al., 2018 [↗](#); Friedman et al., 2024 [↗](#); Galouzis & Furlong, 2022 [↗](#); Zabidi et al., 2015 [↗](#)). Due to such specificity, gene transcription occurs in waves of early and late responsive genes (Fowler et al., 2011 [↗](#); Yamamoto & Alberts, 1976 [↗](#)). Often, the protein factors translated from early genes, regulate the expression of late responsive genes (Dixon et al., 1996 [↗](#); Freter et al., 1996 [↗](#); Herschman, 1991 [↗](#); Williams et al., 1999 [↗](#); Winkles, 1997 [↗](#); Winston & Pledger, 1993 [↗](#)). However, it is not known if the genes that are activated early in signalling time course are spatially related to late activating genes. Further, if spatial proximity of any gene to an early gene is enough to cause the temporal differential transcription due to sequestration of the transcriptional machinery from late gene to early gene is poorly understood. Transcription factors (TFs), after binding to cognate DNA motif, recruit RNA polymerase machinery and co-activators for gene activation (Levine & Tjian, 2003 [↗](#); Liu et al., 2014 [↗](#)). These machinery are limited in supply, and may in effect be sequestered from other genomic regions (Koşar & Erbaş, 2022 [↗](#)). Further, acute activation of genes is linked with phase-separation of TFs, polymerases, mediators and other co-factors/activators, most of which harbour low complexity regions to promote weak protein-protein interactions driving formation of TF-condensates (Boehning et al., 2018 [↗](#); Boija et al., 2018 [↗](#); Cai et al., 2019 [↗](#); Cho et al., 2018 [↗](#); Chong et al., 2018 [↗](#), 2022 [↗](#); Mann & Notani, 2023 [↗](#); Sabari et al., 2018 [↗](#); Shrinivas et al., 2019 [↗](#); Stortz et al., 2020 [↗](#), 2024 [↗](#)). Though the precise stoichiometry of protein molecules in these condensates is unknown, such structures do involve multiple molecules of each transcriptional protein. The formation of such phase-separated compartments potentially can act as a sink for transcriptional machinery depriving neighbouring promoters and enhancers that are not part of the compartment. Such sequestration would cause indirect suppression of these genes that are spatially proximal in the same TAD but not the part of the same condensate.

In order to investigate the effect of an enhancer: promoter pair on a neighbouring gene within the same TAD, we looked at the paradigmatic model of estrogen signalling in mammary epithelial cells namely MCF7 (Levenson & Jordan, 1997 [↗](#); Masiakowski et al., 1982 [↗](#)). We selected E2-induced E-P pair *TFF1* and its neighbouring gene *TFF3* on Chr21 located at a distance of 43kb from each other, as discussed below (Chinery et al., 1996 [↗](#)). We chose to answer these questions using an integrated approach consisting of single molecule RNA FISH (smFISH), genome-wide conformation capture and sequencing (4C-seq), and perturbation of cis-acting regulatory elements by CRISPR. E2-signalling can be induced within minutes by treating the cells with estradiol (E2) which causes activation and repression of several genes across the genome. The peak of E2-mediated signalling occurs at 40 minutes post-induction and starts to decay by 160 minutes (Hah et al., 2011 [↗](#)). This occurs via the binding of estrogen receptor-alpha (ERα) on the enhancers (Li et al., 2013 [↗](#)). *TFF1* is found in a topologically associated domain (TAD) along with a few other genes including *TFF2*, *TFF3*, *TMPRSS3*, and *UBASH3A* (Oh et al., 2021 [↗](#); Quintin et al., 2014 [↗](#); Rao et al., 2014 [↗](#)). *TFF1*

and its neighbouring gene *TFF3* on Chr21 located at a distance of 43kb from each other, as discussed below (Chinery et al., 1996 [\[1\]](#)). We chose to answer these questions using an integrated approach consisting of single molecule RNA FISH (smFISH), genome-wide conformation capture and sequencing (4C-seq), and perturbation of cis-acting regulatory elements by CRISPR. E2-signalling can be induced within minutes by treating the cells with estradiol (E2) which causes activation and repression of several genes across the genome. The peak of E2-mediated signalling occurs at 40 minutes post-induction and starts to decay by 160 minutes (Hah et al., 2011 [\[2\]](#)). This occurs via the binding of estrogen receptor-alpha (ER α) on the enhancers (Li et al., 2013 [\[3\]](#)). *TFF1* is found in a topologically associated domain (TAD) along with a few other genes including *TFF2*, *TFF3*, *TMPRSS3*, and *UBASH3A* (Oh et al., 2021 [\[4\]](#); Quintin et al., 2014 [\[5\]](#); Rao et al., 2014 [\[6\]](#)). *TFF1* and *TFF3* are located at 10kb and 60kb from enhancer respectively (Fig. 1A [\[7\]](#)) and this locus has been useful in answering the questions about promoter-enhancer interactions and multi-gene regulation (Oh et al., 2021 [\[4\]](#); Quintin et al., 2014 [\[5\]](#)).

Using this system, we identified that *TFF1* and *TFF3* genes located within the same TAD show distinct and opposing expression profiles over the course of E2-mediated signalling. In agreement with this, the enhancer also showed increased looping interaction with the cognate gene promoter during peak expression at 1h of signalling. We also identified the role of TF-mediated LLPS on the regulation of these two genes during signalling. Hence, we propose that ligand dependent condensation of ER α can support the expression of the cognate gene in concert with the enhancer while negatively affecting the expression of a neighbouring gene.

Results

ER α binding and ligand-induced gene expression of *TFF1* and *TFF3* change over the course of estrogen signaling

In order to understand how acute activation of one gene in a TAD affects the transcription of a proximate gene, we chose to study *TFF1* and its neighbouring gene *TFF3* at a 43kb distance within same TAD. *TFF1* expression is linked with an enhancer located 10 kb downstream (Li et al., 2013 [\[3\]](#); Saravanan et al., 2020 [\[8\]](#); Oh et al., 2021 [\[4\]](#)) and the distance between *TFF1* enhancer and *TFF3* gene is 53kb (Fig. 1A [\[7\]](#)).

17- β -oestradiol exposure leads to ER α binding on regulatory regions leading to gene activation. Gene transcription of E2-regulated genes was shown to peak at 1h and significantly reduces at 3h due to the rapid degradation of ER α (Hah et al., 2011 [\[2\]](#)). The binding of ER α in the genome also follows this temporal kinetics where it peaks at 1h and reduces at 3h post ligand stimulation (Hah et al., 2011 [\[2\]](#); Li et al., 2013 [\[3\]](#); Liu et al., 2014 [\[9\]](#)). At the *TFF1* locus, the *TFF1* enhancer and to some extent, its promoter, were bound by ER α even in the absence of E2 induction, while it increased in strength at these regions and also at various other regions in the locus at 1h of ligand stimulation (Saravanan et al., 2020 [\[8\]](#)). Notably, the binding at these regions substantially decreased at 3h. On the other hand, ER α did not bind on *TFF3* promoter throughout the course of signalling (Fig. 1A [\[7\]](#) and Fig. S1B [\[10\]](#)). We have previously shown that such estrogen-induced clustered binding of ER α on *TFF1* region is associated with its acute activation post estrogen stimulation (Saravanan et al., 2020 [\[8\]](#)). We then tested the level of their expression by nascent RNA-seq (GRO-seq). We observed a dramatic transcriptional activation of *TFF1* at 1h which reduced substantially at 3h. The expression level of *TFF3* was comparatively low and did not fluctuate upon signalling (Fig. 1A [\[7\]](#)). However, qRT-PCR on unspliced *TFF3* exhibited upregulation at 1h and down-regulation at 3h. (Fig. 1B [\[11\]](#)) While this broad trend is similar to *TFF1* as shown by both GRO-seq and qRT-PCR, the relative changes for *TFF3* was smaller than *TFF1*. We reasoned that these differences in *TFF3* could be because of its low baseline expression (Clark et al., 2015 [\[12\]](#); Conesa et al., 2016 [\[13\]](#); Sha et al., 2015 [\[14\]](#); Svensson et al., 2017 [\[15\]](#); Tarazona et al., 2011 [\[16\]](#)), and lower yet expression of unspliced transcripts. Therefore, in order to test the relative induction of

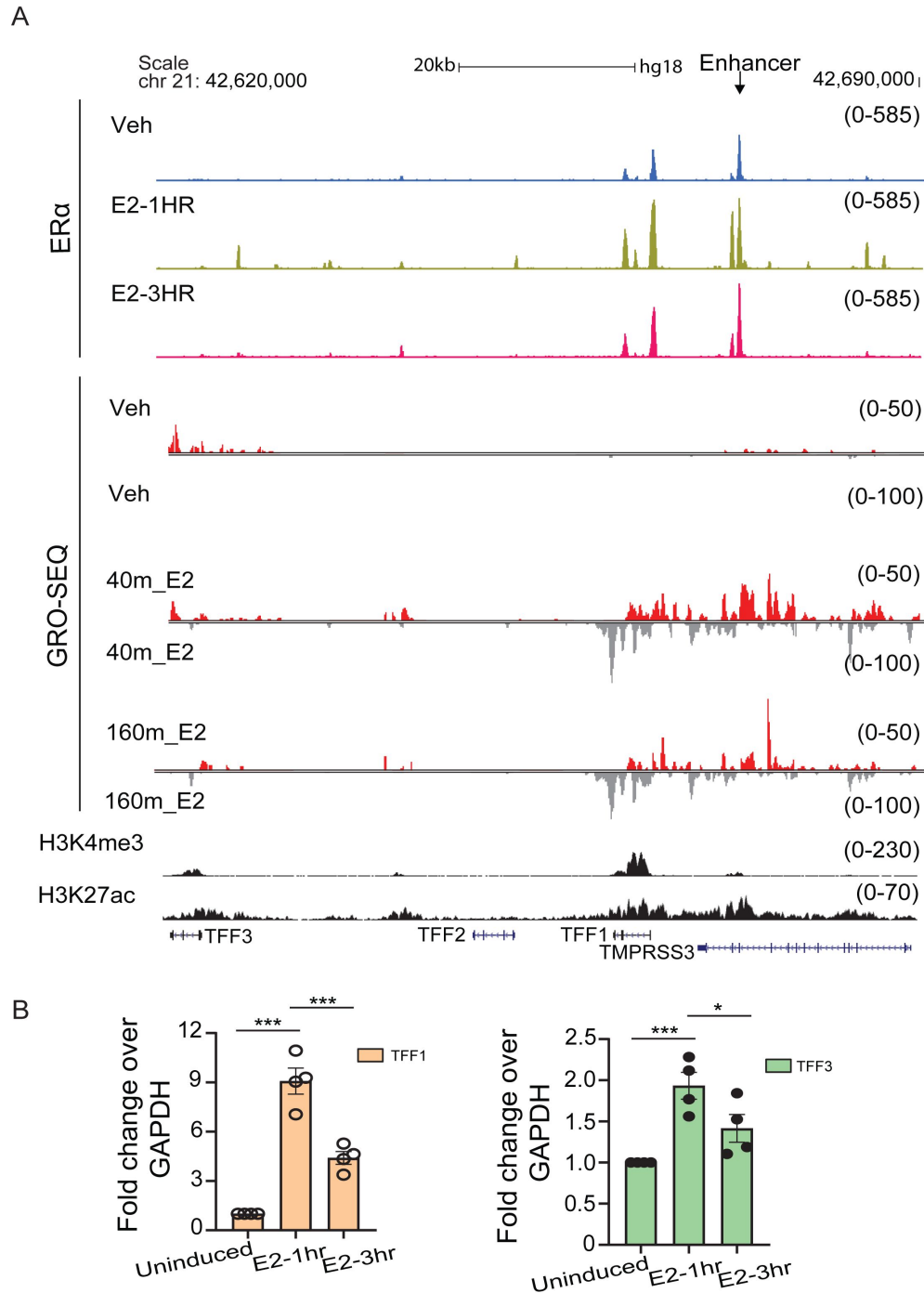


Figure 1.

ERα binding and ligand-induced gene expression of *TFF1* and *TFF3* change over the course of estrogen signaling.

Schematic depicting TFF1 locus, UCSC genome browser snapshots showing the binding of ERα, H3K27ac status, H3K4me3 signal, and Gro-seq signal for robustly E2-induced TFF1 locus. First, second and third ERα ChIP-seq and Gro-seq tracks are from vehicle-treated, E2-1h and E2-3h in WT cells, respectively.

B. qRT-PCR showing the changes in expression of TFF1 and TFF3 genes during the E2 signaling time course. Error bars denote SEM from four biological replicates. Each dot represents a replicate. p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

these genes and their co-regulation, we decided to perform single cell measurements of RNA (**Fig. S1A**) that do not rely on PCR amplification (sc-RNA-seq) and can provide absolute numbers of *TFF1* and *TFF3* transcripts in a given cell. Towards this, we chose to employ smFISH (Femino et al., 1998; Haimovich & Gerst, 2018; Kwon, 2013; Raj et al., 2008).

***TFF1* and *TFF3* exhibit opposite trends during the E2 signalling time-course**

MCF7 cells are hypertriploid to hypotetraploid, and each of the three alleles within a nucleus can behave differently, and smFISH is pre-eminently suited to capturing cell-and allele-specific heterogeneity of expression even for low expressing genes, compared to bulk studies that average over cell populations. Briefly, smFISH allows to visualize single RNA molecules in fixed cells using multiple fluorescently labelled oligonucleotide probes targeted to the RNA of interest and therefore, it can be used to image the transcription and the localization of multiple gene transcripts at same and different time points after signalling (Femino et al., 1998; Raj et al., 2008).

We designed the smFISH probes targeting the intronic region of *TFF1* and *TFF3* to measure their nascent transcripts (referred to as InTFF1 and InTFF3 hereon) whereas the exonic probes were used to primarily measure mature mRNA transcripts (referred to as ExTFF1 and ExTFF3 hereon). Intronic probes are particularly suited for investigating nascent transcriptional status at the time of fixation, while probes against mature mRNA reflects on more steady-state levels of functional mRNA due to finite mRNA lifetimes (Skinner et al., 2016). Additionally, the intronic probes can help determine the localisation and number of alleles that are transcribing at a given time thus allowing for additional interpretations regarding the expression of multiple genes (Skinner et al., 2016).

Representative images from the smFISH experiment probing InTFF1 and InTFF3 are depicted (**Fig. 2A**). The smaller foci represent individual intronic/nascent transcripts while the larger foci represent the site of transcription (Raj et al., 2006; Zenklusen et al., 2008). Since MCF7 cells are hypertriploid in nature, we expected to see 1-3 large foci per cell representing sites of transcription. Additionally, we observed many individual transcripts labelled by intronic probes (**Fig. 2A, D**) within the same nuclei (probes against mature mRNA are usually more cytoplasmic). This is suggestive of the fact that the transcripts undergo non-co-transcriptional splicing as the transcripts labelled by intronic probes are localised away from the site of transcription. This is not surprising as several reports have shown that nascent transcript can undergo splicing well after transcription (Coulon et al., 2014; Drexler et al., 2020; Khodor et al., 2012). Indeed, a very recent study has shown that such post-transcriptional splicing occurs in a small zone around genes including *TFF1* (Coté et al., 2023). Therefore, observation of nascent transcripts away from the site of transcription as we see for *TFF1* and *TFF3* (**Fig. 2A, D**) is not unexpected. We quantified the number of such transcripts per nuclei as a proxy for ongoing transcription.

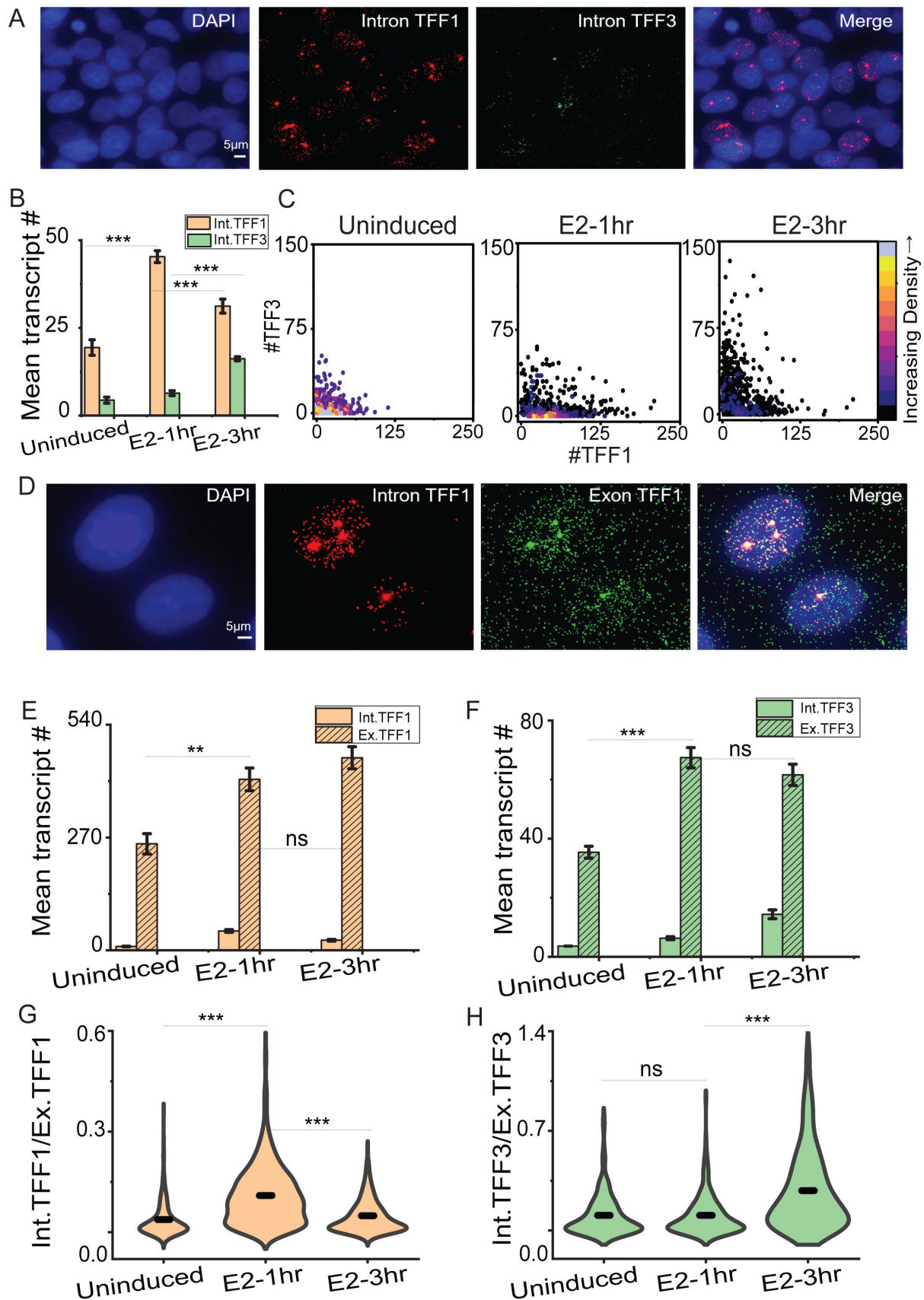


Figure 2.

TFF1 and TFF3 expressions show opposite trends during the E2 signaling time-course

A. 60X Representative images from single molecule RNA FISH experiment showing transcripts for TFF1 and TFF3. The probe was designed against the unspliced RNA containing the intronic region. The scale bar is 5 microns.

B. The mean RNA numbers are depicted. These are counted using an in-house MATLAB code which uses the DAPI-stained nuclei as the mask to count the RNA present in the nucleus. The graph shows the mean of means from three different repeats of the experiment, and error bars denote SEM ($n = 665$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

C. Scatter plots showing the distribution of InTFF1 and InTFF3 on a cell-by-cell basis ($n = 665$, $N = 3$). The absolute RNA numbers are combined from three different repeats. Density plots have been used to clearly visualize overlapping data points.

D. 60X Representative images from single molecule RNA FISH experiment showing transcripts for InTFF1 and ExTFF1. Scale bar is 5 microns.

E. The mean RNA numbers for InTFF1 and ExTFF1 are depicted. Separate probes were used to target unspliced (InTFF1) and mature (ExTFF1) RNA. These are counted using an in-house MATLAB code which uses the DAPI-stained nuclei as the mask to count the intronic RNA present in the nucleus and a free-drawn region to designate the cell to count the exonic RNA present in the nucleus as well as the cytoplasm. The graph shows the mean of means from three different repeats of the experiment, and error bars denote SEM ($n > 360$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

F. The mean RNA numbers for InTFF3 and ExTFF3 are depicted. Separate probes were used to target unspliced (InTFF3) and mature (ExTFF3) RNA. The graph shows the mean of means from three different repeats of the experiment, and error bars denote SEM ($n > 210$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

G. Violin plots showing the ratio of intronic to exonic TFF1 counts are depicted. The graph shows the distribution of ratios combined from three different repeats ($n > 360$, $N = 3$). p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

H. Violin plots showing the ratio of intronic to exonic TFF3 counts are depicted. The graph shows the distribution of ratios combined from three different repeats ($n > 210$, $N = 3$). p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

At 1h post E2 induction, *TFF1* mean transcript counts increased significantly compared to uninduced, whereas, the increment was less pronounced for *TFF3* transcripts. In contrast, the mean transcript counts for *TFF3* increased significantly at 3h post induction while *TFF1* transcription showed a decrease at 3h compared to 1h (Fig. 2B). To check the transcriptional status of *TFF1* and *TFF3* in the same cell, we plotted the transcript counts from individual cells (Fig. 2C). The data suggested that the transcript counts for *TFF1* increased at 1h and cells that showed high counts for *TFF1* were likely to have low counts for *TFF3*. But the RNA counts for *TFF1* decreased at 3h and increased for *TFF3*. Overall, it was evident that the transcriptional profile for these two genes located in the same TAD were negatively correlated as they peaked at different time points. To further confirm this, smFISH using probes targeting both the intronic and exonic

transcripts in the same experiment was conducted. Intronic probes represent active transcription, while exonic probes show accumulated mature mRNA from past transcriptional events even in the absence of active transcription (Skinner et al., 2016 [DOI](#)). Representative images from the smFISH experiment probing InTFF1 and ExTFF1 are depicted (**Fig. 2D** [DOI](#)). We observed that the exonic transcript counts for both *TFF1* and *TFF3* increased at 1h compared to uninduced (**Fig. 2E-F** [DOI](#)). Strikingly, at 3h, the exonic transcripts for *TFF1* continued to increase even while the intronic transcript counts reduced, though the fold increase (1.12 ± 0.02) was less compared to that between uninduced and 1h (1.63 ± 0.28), indicating a plateauing of steady state levels. Exonic transcripts for *TFF3* at 3h remained comparable to 1h while the intronic counts increased suggesting, *TFF1* transcription reduces at 3h whereas, *TFF3* expression increases. We reasoned that the ratio of intronic transcript number (InTFF1) to exonic transcripts number (ExTFF) should represent the status of transcription as an increase in the number of intronic transcript due to expression would result in a higher ratio while also taking into consideration the number of mature transcripts. As expected, the ratio of intronic transcripts to exonic transcripts also showed that transcription is active at 1h for *TFF1* as the ratio is higher compared to uninduced and 3h (**Fig. 2G** [DOI](#) and **Fig. S2A** [DOI](#)). Contrastingly, the ratio was highest at 3h for *TFF3* suggesting that active transcription takes place much after the peak of E2-mediated signalling and maximal *TFF1* expression (**Fig. 2H** [DOI](#) and **Fig. S2B** [DOI](#)).

***TFF1* enhancer does not change target promoters during signalling time-course**

In order to identify the molecular players behind differential expression of these two genes that are in the same TAD, we asked if the ER α -bound enhancer downstream to *TFF1* gene, loops with *TFF1* at 1h, and with *TFF3* at 3h. This enhancer acutely activates *TFF1* at 1h post estrogen stimulation (Saravanan et al., 2020 [DOI](#); Ho et al., 2021). We interrogated the looping using enhancer as a viewpoint by 4C-seq in uninduced, 1 and 3h post E2 stimulation. We observed robust interactions between enhancer and *TFF1* promoter at 1h post induction which reduced at 3h. On the other hand, its interaction with *TFF3* promoter exhibited very low counts in uninduced as well as E2-induced conditions at both time points and in both replicates (**Fig. 3A** [DOI](#), Fig. S3A). This suggests that the looping of enhancer was potentially inducing the expression of *TFF1* gene at 1h and loss of interactions at 3h resulted in weak *TFF1* transcription. However, lack of interactions between enhancer and *TFF3* did not explain the gain of *TFF3* expression at 3h post ligand stimulation.

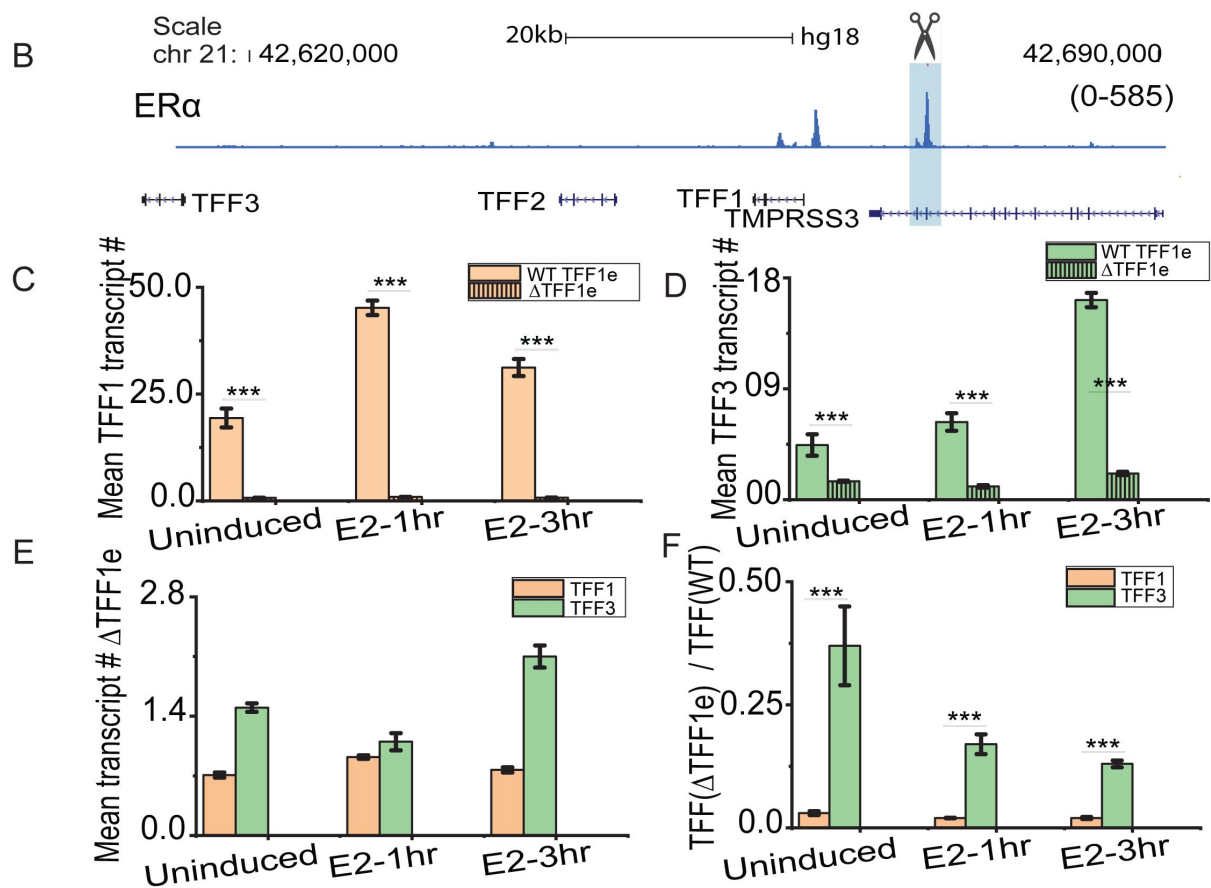
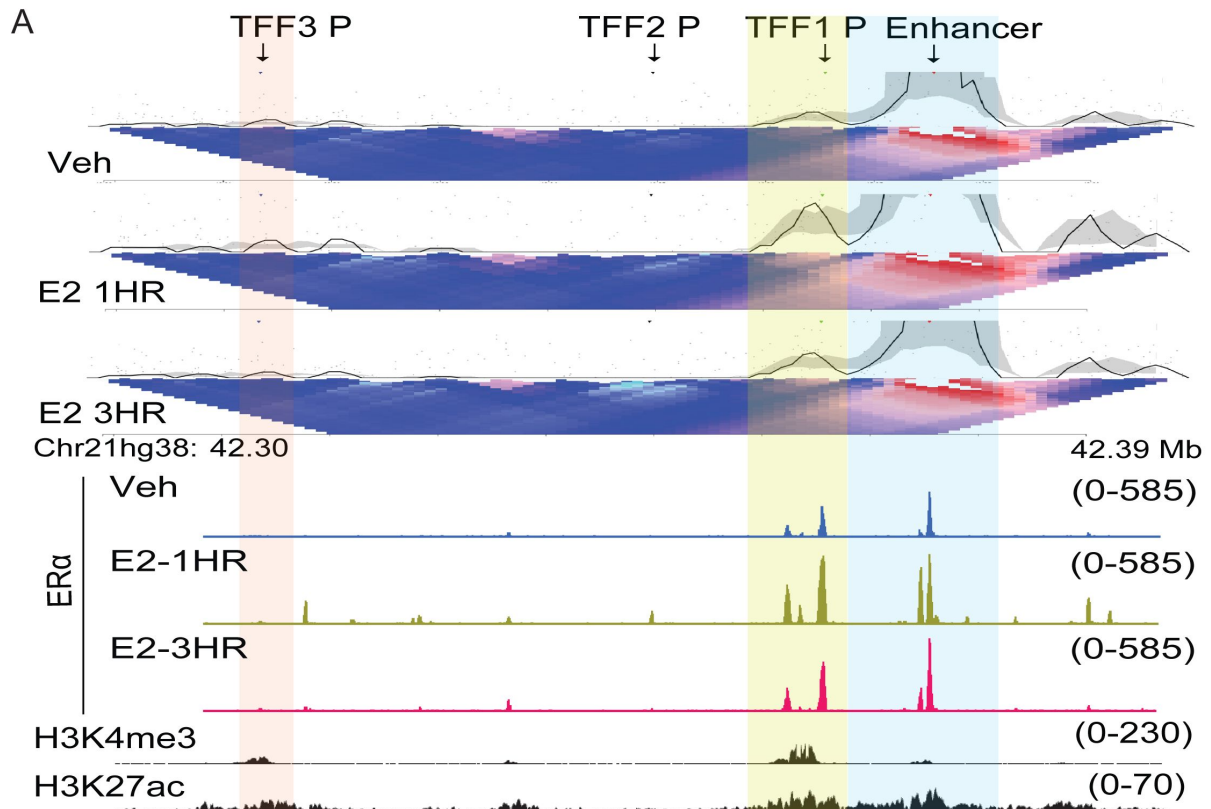


Figure 3.

Enhancer looping does not account for the differential expression of *TFF1* and *TFF3* genes

A. 4C-seq plot at *TFF1* enhancer viewpoint, the interaction with the promoter is highlighted in yellow. The plot is overlaid with H3K27ac, ERα ChIP signal, and gene annotations.

B. Genome browser snapshot of *TFF1* region depicting ERα binding in WT lines. The first, second and third ERα ChIP-seq tracks are from WT cells that are vehicle-treated, E2-1h, and E2-3h, respectively. Blue highlighted regions represent the $\Delta TFF1e$ region.

C. The mean RNA numbers for InTFF1 in WT (unshaded) and $\Delta TFF1e$ (shaded) MCF7 cells are depicted. The mean of means are shown, and error bars denote SEM from three repeats ($n > 650$, $N = 3$ for each WT and delete line). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

D. The mean RNA numbers for InTFF3 in WT (unshaded) and $\Delta TFF1e$ (shaded) MCF7 cells are depicted. The mean of means are shown, and error bars denote SEM from three repeats ($n > 650$, $N = 3$ for each WT and delete line). P-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

E. The mean RNA numbers for InTFF1 and InTFF3 in $\Delta TFF1e$ MCF7 cells are depicted. The mean of means are shown, and error bars denote SEM from three repeats ($n > 880$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

F. Ratio of InTFF in WT MCF7 to InTFF in $\Delta TFF1e$ MCF7 are depicted. The ratio was obtained by dividing the absolute RNA counts of the WT line by delete lines performed on different days but in the same order (replicate one of WT divided by replicate one of $\Delta TFF1e$). The mean of means are shown, and error bars denote SEM from three repeats ($n > 650$, $N = 3$ for each WT and delete line). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

This could mean that enhancer interaction is critical for *TFF1* expression, but is less important for *TFF3* expression. Thus, *TFF1* expression should be affected more severely upon deletion of the enhancer than *TFF3*. To test this, we investigated the expression of *TFF1* and *TFF3* in MCF-7 cells where the enhancer downstream to *TFF1* was homozygously deleted using CRISPR-Cas9 (referred to as the $\Delta TFF1e$ from hereon) (Saravanan et al., 2021). Using smFISH, we quantified the intronic transcripts for *TFF1* and *TFF3* in the $\Delta TFF1e$ cells, compared to WT cells. We observed that the mean number of *TFF1* transcripts was reduced drastically in the $\Delta TFF1e$ compared to the WT (Fig. 3C). The reduction was far more substantial (49.23 ± 2.6 at 1hr and 40.75 ± 7.8 at 3h) for *TFF1* than *TFF3* (6.03 ± 1.8 at 1hr and 7.6 ± 0.7 at 3h) (Fig. 3D). The absolute transcript counts for *TFF1* and *TFF3* in the $\Delta TFF1e$ cells have also been shown for clearer visualization (Fig. 3E), as these are obscured when compared to WT. To get a sense of fold changes, we took a ratio of mean transcript counts in WT cells to $\Delta TFF1e$ cells at each time point. We observed that the drop in gene transcripts between WT and $\Delta TFF1e$ were several folds higher for *TFF1* compared to *TFF3* (Fig. 3F). This suggests that enhancer deletion has a more robust impact on the transcription of *TFF1* compared to *TFF3*. Nonetheless, *TFF3* was also affected even though, it does not loop with the enhancer. This is in accordance with the 4C-seq data where we observed prominent looping between the enhancer and *TFF1* but less so with *TFF3*. These results suggest that the enhancer plays a more important role in the expression of the primary gene while it has less impact on a gene located more distally but within the same TAD during course of signalling.

Levels of ERα in the nucleus dictate the extent of *TFF1* and *TFF3* inductions

After ruling out enhancer looping as the determinant of differential expression, we looked for other candidates that could regulate the differential gene expression. As discussed above, globally, E2-mediated gene expression is known to peak at 1h after stimulation and drop significantly by 3h (Hah et al., 2011 [DOI](#)). Upon ligand stimulation, ERα translocates into nucleus, increasing mean ERα intensity in the nucleus which is high at 1h and then decreases significantly by 3h due to degradation. These changes in intensities have been captured using immunofluorescence for ERα (Saravanan et al., 2020 [DOI](#)). We tested if the intensities of ERα in individual cells correlate with the expression of *TFF1* and *TFF3* at 1h and 3h of E2 signalling. Towards this, we combined smFISH with immunofluorescence for ERα. To improve contrast for the ERα signal, we additionally performed a chromatin retention assay to get rid of any chromatin unbound ERα (**Fig. 4A** [DOI](#)). The representative images show that the cells with very high levels of nuclear ERα (blue circle) exhibited low counts of both *TFF1* and *TFF3* while the cells with medium levels of ERα (red circle) possessed higher *TFF1* than *TFF3*. Similarly, the cells with the lower levels of ERα (grey circle) showed higher *TFF3* expression as compared to *TFF1* (**Fig. 4A** [DOI](#)). Histograms depicting the ERα mean intensities across individual cells, showed that the nuclear level of ERα increases post 1h of induction and then goes down at 3h (**Fig. 4B, C** [DOI](#)), similar to *TFF1* expression (**Fig. 4D** [DOI](#)).

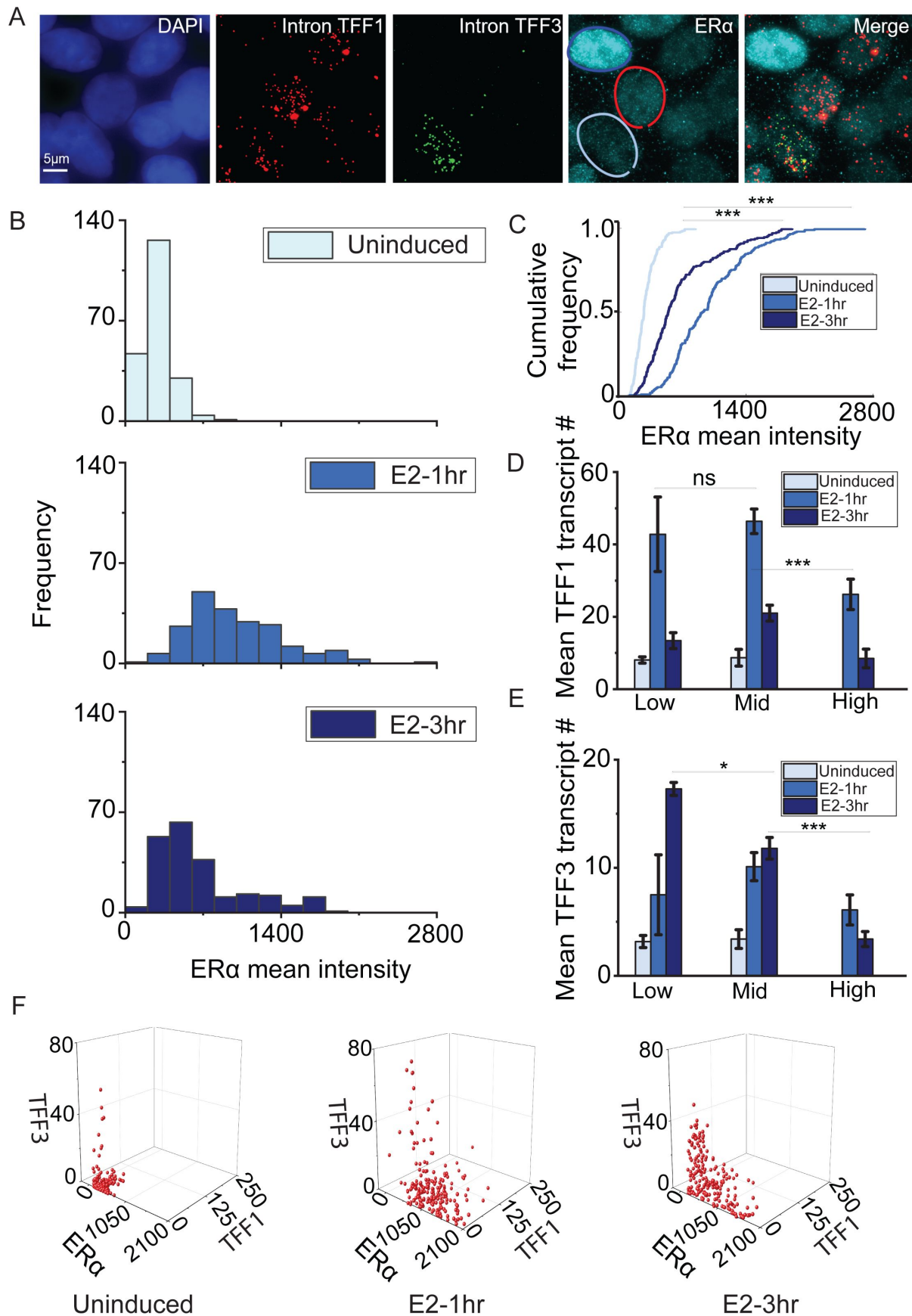


Figure 4.

Nuclear levels of the transcription factor ERα dictate the extent of expression of TFF1 and TFF3 genes

A. Representative images showing smFISH for InTFF1 and InTFF3 in combination with immunofluorescence for ERα (along with chromatin retention assay). The blue circle denotes a cell with high ERα and low TFF1 and TFF3, the red circle denotes a cell with medium ERα, high TFF1, and low TFF3, while the cyan circle denotes a cell with low ERα and high TFF3. The scale bar is 5 microns.

B. Histogram representing the distribution of ERα mean intensities in cells under induced, E3-1hr, and E2-3hr conditions (n=210). Intensities at 1hr are the highest while they shift to the left at 3hr and are lowest in uninduced cells. This is plotted from one experimental repeat out of three repeats, as ERα intensities will vary from one immunofluorescence experiment to another.

C. Cumulative histogram representing the distribution of ERα mean intensities in cells under induced, E3-1hr, and E2-3hr conditions (n=210). p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

D. ERα intensities were sorted into three categories, namely low (intensities between 0-450 A.U.), mid (intensities between 450-1200 A.U.) and high (intensities between 1200-2100 A.U.). The mean and SEM of transcript count for InTFF1 in the three categories under uninduced, E2-1hr, and E2-3hr were plotted (n=210). Low and mid categories show the highest TFF1 mean. p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, and ns denotes $p > 0.05$. This is plotted from one experimental repeat out of three repeats, as ERα intensities will vary from one immunofluorescence experiment to another.

E. ERα intensities were sorted into three categories, namely low (intensities between 0-450 A.U.), mid (intensities between 450-1200 A.U.), and high (intensities between 1200-2100 A.U.). The mean and SEM of transcript count for InTFF3 in the three categories under uninduced, E2-1hr, and E2-3hr were plotted. The low category shows the highest TFF3 mean. p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$. This is plotted from one experimental repeat out of three repeats, as ERα intensities will vary from one immunofluorescence experiment to another.

F. 3D plot representing the distribution of ERα, InTFF1, and InTFF3 on a cell-by-cell basis shows that cells with lower levels of ERα show higher counts for InTFF3. This is plotted from one experimental repeat out of three repeats, as ERα intensities will vary from one immunofluorescence experiment to another (n=210).

To further corroborate this, we parsed the ERα population cells into 3 categories namely low, medium, and high within the cells imaged at 1 and 3h. We plotted the mean counts of *TFF1* in each of these bins at different time points and observed that the mean count was higher in the mid-category (Fig. 4D). While for *TFF3*, the mean count was significantly higher in the low bin (Fig. 4E). The transcript counts for *TFF1* and *TFF3* against ERα intensities on a cell-by-cell basis (Fig. 4F), also showed this feature where very high levels of ERα in the nucleus were not conducive to the expression of either gene (Fig. 4F). To test if ERα had a causal role in intensity based expression of *TFF1* and *TFF3*, we increased the levels of ERα by over expression of ERα-GFP (Fig. S4A,C). The representative images from smFISH experiment in cells overexpressing ERα-GFP confirmed that *TFF1* and *TFF3* were downregulated in transfected cells while these genes were not perturbed in non-transfected cells in neighbourhood (Fig. S4A, C). Meanwhile, the cells overexpressing ERα-GFP do not show any impairment in the expression of GAPDH (housekeeping gene) (Fig. S4A,B). As another control, we transfected the cells with EGFP-C1 (same backbone as ERα-GFP construct) and observed no effect on *TFF1* or *TFF3* (Fig. S4D,E). The data suggest that

loss of *TFF1* and *TFF3* expression upon ER α overexpression was not a general effect of transfection stress, but rather specific to ER α overexpression. These results indicate that high nuclear levels of ER α can be detrimental to the expression of genes it regulates. This could be due to widespread condensate formation which in turn could sequester the transcriptional protein complexes and competitively abrogate transcription across the multiple loci. Thus, the global level of ER α in the nucleus can be predictive of transcriptional status of specific genes. The binding of ER α at 1 and 3h is proportional to its nuclear levels (**Fig. 3A**); suggesting, its overexpression would lead to more binding in the genome which is detrimental to gene expression.

LLPS perturbation down-regulates

TFF1 but supports *TFF3* expression

The data obtained from combined smFISH and ER α immunofluorescence indicates that ER α could be a determining factor in controlling the differential gene expression of *TFF1* and *TFF3*. Existing literature indicates that ER α -mediated condensate plays a role in E2-induced gene expression (Boija et al., 2018; Nair et al., 2019; Sabari et al., 2018; Saravanan et al., 2020). It led us to hypothesize that ER α -mediated condensate at the *TFF1* locus could be sequestering all the factors required for active transcription and thus preventing activation of the *TFF3* locus at the active phase of E2 signalling. To validate this, we treated the cells with 3% 1,6-Hexanediol for 5 minutes which is known to disrupt LLPS (Gamliel et al., 2022; Kroschwald et al., 2017). Following the treatment, we performed smFISH to look at the transcription of *TFF1* and *TFF3* in the same cell and observed a dramatic reduction in *TFF1* transcripts, whereas statistically significant increase in *TFF3* was noted (**Fig. 5A**). Since, ER α forms condensates only after estrogen stimulation, 1,6-Hexanediol had no effect on *TFF1* in the absence of E2 signalling (**Fig. S5A**). Together, these results suggest that the functional loss of *TFF1* promoter transcription due to the dissolution of ER α condensate allowed *TFF3* promoter in the neighbourhood to gain access to transcriptional machinery leading to its upregulation. In order to test the generality of this observation, beyond the *TFF1*, *TFF3* locus, we divided E2-responsive genes in three categories, low, moderate, and high based on their expression upon E2-40m stimulation. We observed the significant increase at 40m and down-regulation at 160m when the binding of ER α is reduced in the genome (**Fig. 5B**). Next, we tested the expression of their nearby upregulated genes at 160m compared to 40m post E2 stimulation. The nearby genes are within a distance of 1Mb from the E2-responsive genes which is approximately the average size of a TAD. Indeed, we observed the significant up regulation of nearby genes at 160m when the expression of highly induced genes dropped. Additionally, the expression of these genes was reduced at the peak of signalling (**Fig. 5B**), showing that at 40m of signalling, the acute activation of primary genes and sequestration of transcription machinery by these genes leads to the loss of expression of nearby genes.

Discussion

smFISH allowed us to simultaneously capture allele level transcription of two genes in the same TAD, *TFF1* and *TFF3* at single cell level during the peak and fall of signalling. We were able to capture the anti-correlated expression of these two genes revealing an intricate regulatory feedback between acutely activated enhancer-dependent gene, *TFF1* which caused the dysregulation of non-enhancer targeted gene, *TFF3* within the same TAD (**Fig. 5B**).

Our data suggests that while condensate formation on enhancer allows robust activation of target gene however, it negatively impacts the expression of other neighbouring genes potentially due to the sequestration of transcriptional machinery from these genes. When the condensates dissolve, the locally enriched transcriptional machinery is available to the other loci, potentially allowing for increased transcription of the these genes, that are not the direct target of enhancer (**Fig 5C**). Forced dissolution of condensates allows the expression of non-enhancer target genes in the cells

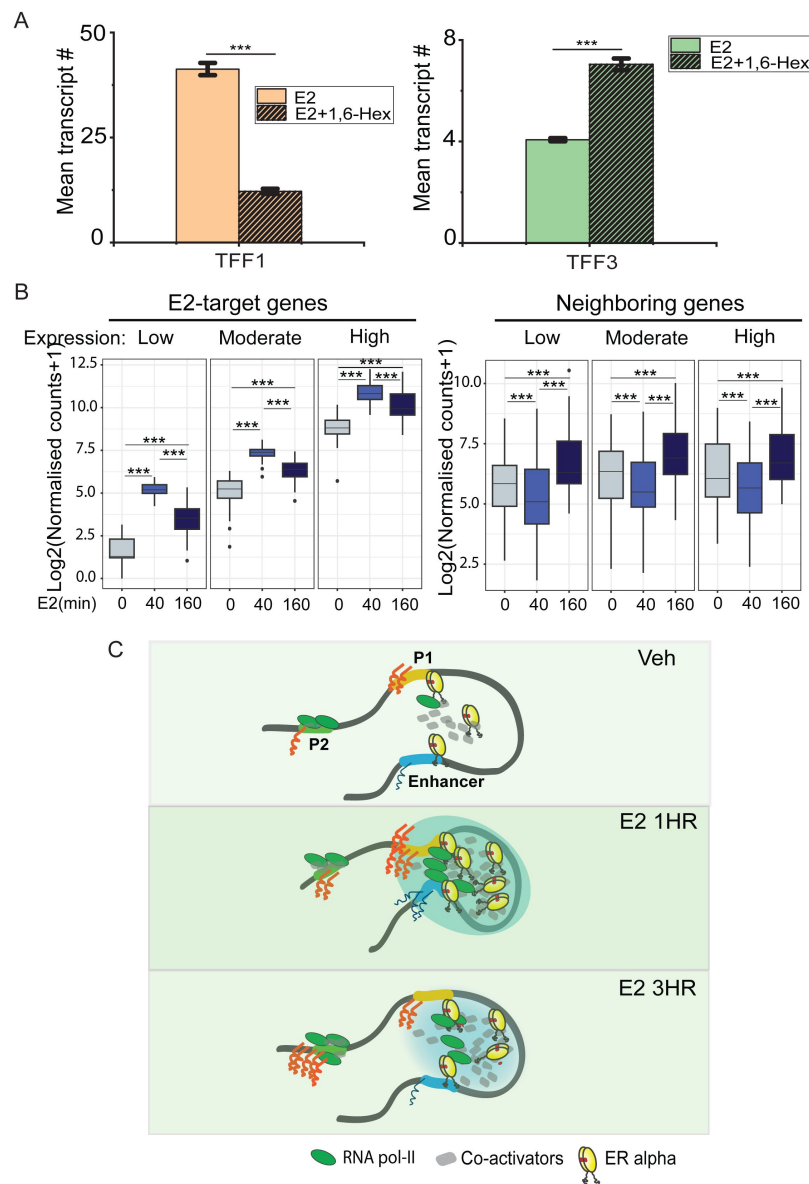


Figure 5.

ERα mediated phase separation could govern the differential expression of TFF1 and TFF3 genes

A. Mean transcript counts for InTFF1 and InTFF3 in control, and 3% 1,6-Hexanediol treated cells post 30 minutes of E2-induction. The mean of means are shown, and the error bars denote SEM from three repeats ($n > 880$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

B. Boxplots showing DESeq2 normalized counts for low expressing, moderately expressing, and highly-expressing genes in the vehicle, E2-40m, and E2-160m respectively (left). Boxplots showing DESeq2 normalized counts for genes near low expressing, moderately expressing, and highly expressing genes in the vehicle, E2-40m, and E2-160m, respectively (right). The p-values in the boxplots were calculated using the Wilcoxon rank-sum test. The boxplots depict the minimum, first quartile, median, third quartile, and maximum values, along with outliers.

C. Model depicting the signaling under uninduced, E2-1hr, and E2-3h conditions-1. Activation of TFF1 locus by ligand induction. 2. During the active phase, liganded ERα binds on enhancer and promoter. Together these in 3D interact, manifesting as ERα puncta, which results in robust expression of target genes, 3. The phase-separated condensate leads to the sequestration of transcriptional machinery. 4. This favours intra-TAD enhancer: promoter interactions to facilitate robust gene transcription. Upon ERα degradation at 3h, these condensates disappear, and transcriptional machinery becomes available to other promoters leading to their somewhat increased gene transcription.

although at low levels. The data also suggests that globally, the neighbouring genes to ligand induced genes are upregulated at the fall of signalling as an indirect consequence of excess polymerase availability in the neighbourhood.

Non-enhancer target genes are also regulated indirectly in the same TAD

ER α binding strength increases at the enhancer of *TFF1* at 1h of ligand stimulation (Saravanan et al., 2020 [↗](#)). Such estrogen-induced binding of ER α leads to robust ligand induced activation of *TFF1* gene. While, as *TFF3* promoter remains unbound by ER α (Fig. 3A [↗](#)) and does not interact with *TFF1* enhancer throughout the course of signalling. These data suggest that *TFF3* expression is both ER α and enhancer independent. Despite non-dependence, *TFF3* expression was mutually exclusive to *TFF1* expression and was dampened in the absence of enhancer. The latter could be due to the enhancer mediated repositioning of the entire TAD that benefitted the expression of *TFF3*.

The mean expression of *TFF1* was many fold higher than *TFF3* thus explaining its dependence on an active enhancer to its upstream. These cells that expressed *TFF1*, did not express *TFF3* as efficiently suggesting Pol2 was potentially sequestered from *TFF3* promoter. However, at 3h when liganded ER α degrades leading to dissolution of condensates on enhancer, *TFF3* expression increased suggesting a local redistribution of active transcription machinery to other genes within the same chromatin domain. Even at 1h, when condensates were perturbed by 1,6 HD, the expression of *TFF3* increased suggesting the access to free Pol2 pool by *TFF3* promoter. The data explains negative-correlation between *TFF1* and *TFF3* expression at the peak of signalling and robust activation of *TFF3* at 3h when *TFF1* expression decreased at single cell level.

High and low levels of ER α were detrimental to enhancer mediated activation of TFF1

The cells with very high nuclear levels of ER α do not support optimum transcription of either gene. Further, *TFF1* expression was more in cells showing mid-ER α levels at 3h while, *TFF3* peaked in cells with low ER α levels. The reason for this observation could lie in the fact that very low levels of TF do not reach or exceed the critical concentration required for LLPS (for *TFF1*) but this poor transcriptional state of *TFF1* was favourable for *TFF3*. Alternatively, a very high concentration of the TF could result in the formation of extremely dense homogenous condensates effectively reducing transcription (Chong et al., 2022 [↗](#); Ryu et al., 2024 [↗](#)). Our data suggests that such a mechanism may be general and not just applicable to the *TFF1*-*TFF3* pair (Fig. 5B [↗](#)). Together, our study underscores the indirect effects of ligand-induced chronic gene transcription that is dependent on enhancer activation and phase-separation within a TAD.

Materials and methods

Cell culture

MCF-7 cells (WT scramble control and $\Delta TFF1$ e were generated in Saravanan et al., 2020 [↗](#)) cells were cultured in non-stripping media consisting of DMEM (Gibco, 12100-046) supplemented with 10% FBS (Gibco, 16000-044) and 1% Penicillin-Streptomycin-Glutamine (Gibco, 10378-016) in a 5% CO₂ humidified incubator at 37 °C (unstripped MCF-7). Cells were passaged and seeded into glass bottom dishes in non-stripping media for 24h and allowed to reach 60–80% confluency. These cells were then hormone-stripped for three days in stripping media containing phenol red-free DMEM (Gibco, 21063-029) supplemented with 5% charcoal-stripped FBS(Gibco,12676029) and maintained in the humidified incubator at 37 °C (stripped MCF-7).

In order to induce the estrogen transcriptional responses, on the third day cells were treated with β -estradiol (E2758, Sigma-Aldrich) at 100nM concentration for various periods as mentioned in the respective Figures. For untreated control, cells were either treated with equal microliters of ethanol on the third day or with ER α inhibitor ICI182780 (1047, Tocris Biosciences) at 100nM concentration for 24 h after two days of stripping.

For perturbation of LLPS in E2-induced condition, cells were incubated with media containing 3% 1,6-Hexanediol for 5 minutes (after 30 minutes of E2 treatment) followed by removal of 1,6-Hexanediol and recovery for 25 mins in E2 containing media.

For experiments with overexpression of EGFP-ER α or EGFP, cells were transfected 24 hours before E2 induction using X-tremeGENE™ HP DNA Transfection Reagent (XTGHP-RO) in stripping media.

Probe for smFISH

Probes were designed, custom-made, and tagged with indicated fluorophores (Quasar 570 or Cal 610) by Biosearch Technologies (<https://www.biosearchtech.com/products/rna-fish>). Probes were made to target the intronic or exonic regions of the *TFF1* and *TFF3* genes. Probe sequences can be found in **supplementary table 1** in supplementary materials.

Chromatin retention assay and smFISH

For smFISH experiments without chromatin retention assay cells, the cells were treated with estradiol or vehicular control for indicated times. Following this, the cells were washed with nuclease-free 1X PBS (Ambion, AM9624) twice. This was followed by fixation using 4% paraformaldehyde (PFA, Sigma, P6148) in 1X PBS for 10 minutes at room temperature. The fixative was removed and cells were washed twice with 1X PBS. The cells were permeabilized using 70% ethanol at 4°C overnight. On the next day, the permeabilizing agent was removed and the cells were washed twice with 1X PBS. The cells were washed once with a wash buffer (10% Formamide (Ambion, 9342) and 2X SSC (Ambion, AM9763) in nuclease-free water) for 5 minutes at room temperature. The wash buffer was aspirated and the cells were incubated with a hybridization mix (100 μ l of mix contains 10% formamide, 89 μ l hybridization buffer, and 1 μ l each of indicated probes) overnight at 37°C in a humid chamber. The next day, the solution was removed and cells were washed with the wash buffer at 37°C for 30 minutes. To stain the nuclei, DAPI (Invitrogen, D1306; 2 μ g/ml) in wash buffer was added to the cells and incubated for 10 minutes at 37°C. The cells were then washed with 2X SSC for 5 minutes at 4°C. The solution was removed and the cells were covered with a few drops of the mountant Vectashield (Vector Labs). The plates were imaged after at least 1h.

For smFISH experiments with chromatin retention assay cells were treated with estradiol or vehicular control for indicated times. Following this, the cells were treated with CSK buffer (consisting of 10 mM PIPES buffer, 100 mM NaCl, 3 mM MgCl₂, 300 mM sucrose, and 0.7% Triton-X 100) for 15 minutes at room temperature. After this, the cells were washed with nuclease-free 1X PBS twice. This was followed by fixation using 4% paraformaldehyde in 1X PBS for 10 minutes at room temperature. The fixative was removed and cells were washed twice with 1X PBS. The cells were permeabilized using 0.3% Triton-X 100 (Sigma, T8787) in 1X PBS for 10 minutes at room temperature. The permeabilizing agent was aspirated and the cells were washed twice with 1X PBS. This was followed by a washing step using the wash buffer for 5 minutes at room temperature. Following this, the cells were incubated with the hybridization mix to which the antibody of interest was also added at appropriate dilution. The next day, the hybridization mix containing probes and antibodies was removed. Cells were washed twice with 1X PBS. Then the cells were incubated with 1X PBS containing a secondary antibody against the antibody of interest at room temperature for 2h. This was followed by incubation with the wash buffer at 37°C for 30 minutes. To stain the nuclei, DAPI in wash buffer was added to the cells and incubated for 10

minutes at 37°C. The cells were then washed with 2X SSC for 5 minutes at 4°C. The solution was removed and the cells were covered with a few drops of the mountant Vectashield. The plates were imaged after at least 1h.

Antibody staining/ immunofluorescence

The primary antibody against ER α (Santa Cruz, sc8002(F10)) was added to the hybridization mix at the dilution of 1:400 and incubated overnight at 37°C. The next day, an anti-mouse Alexa Fluor™ 488 secondary antibody (Invitrogen, A11029) was added to 1X PBS at the dilution of 1:1000. Cells were incubated for 2h followed by the continuation of the smFISH protocol as indicated above.

Image acquisition

The plates were imaged on an Olympus IX83 inverted widefield fluorescence microscope with a Retiga 6000 CCD monochrome camera (QImaging). The images were acquired using a 60X, 1.42 N.A. oil immersion objective or a 100X, 1.4 N.A. oil immersion objective. The z-step size was 0.3 μ m and 35 slices were acquired. The resolution at which the images were acquired is 2752 x 2208. Narrow band-pass filters were used to distinguish the signal from Quasar 570 and Cal 610 labeled probes (ChromaTechnology-49309 and 49310).

Image analysis and representation

The images of mRNA channels were subjected to background subtraction using a rolling ball radius of 10 pixels across the entire stack using Fiji. For representative images, the stacks were Z-projected in Fiji and shown. Transcripts were counted using either Rajlabimagetools (courtesy of Arjun Raj lab) or an in-house MATLAB (Mathworks) script based on earlier work (Femino et al., 1998 [\[4\]](#); Raj et al., 2008 [\[4\]](#)). Briefly, the code uses a nuclear mask to count the intron-containing RNA present in the nucleus (InTFF), and another mask for the whole cell to count the mature mRNA, detected by exonic probes (ExTFF), that are present in the nucleus as well as the cytoplasm. The mean intensity for the ER α immunofluorescence channel was also quantified using the MATLAB script on a cell-by-cell basis using the nuclear mask.

Graphing and Statistics

The graphs were plotted using Python 3, MATLAB, and Origin Pro and edited using Adobe Photoshop. To perform a t-test for data combined from three repeats, GraphPad (<https://www.graphpad.com/quickcalcs/ttest1/> [\[4\]](#)) was used. Non-parametric tests were performed in case of single cell data. For this Mann-Whitney test was used. Significance is represented as : *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

Circular Chromatin Conformation Capture-seq

4C was performed as per the protocol described in van de Werken et al., 2012 [\[4\]](#) with minor variations. MCF7 cells were fixed with fresh formaldehyde (1.5%) and quenched with glycine (125mM) followed by washes with ice-cold 1XPBS (2X) and scraped, pelleted, and stored at -80°C. Lysis buffer [Tris-Cl pH 8.0 (10mM), NaCl (10mM), NP-40 (0.2%), PIC (1X)] was added to the pellets and homogenized by Dounce homogenizer (20 stroked with pestle A followed by pestle B). The 3C digestion was performed with DpnII (200 units, NEB) and ligation was performed by T4 DNA ligase and ligation mix [Triton X-100 (1%), 1x Ligation buffer (10X Ligation buffer-Tris-Cl pH 7.5 (500mM), MgCl₂ (100mM), DTT (100mM), BSA (0.105mg/ml), ATP (1.05mM)]. The ligated samples were purified by PCI and subjected to ethanol precipitation. The pellet was eluted in 1X TE (pH 8.0) to obtain the 3C library. The 4C digestion was performed by NlaIII (50 units, NEB), and the samples were ligated, purified, and precipitated similar to the 3C library to obtain the 4C library. The 4C library was subjected to RNaseA treatment and purified with the QIAquick PCR purification kit. The concentration of the library was then measured by Nanodrop and subjected to PCRs using the

oligos for the enhancer viewpoint. The samples were next PCR purified using the same kit and subjected to next-generation sequencing with Illumina HiSeq2500/ NOVA seq. The 4C oligos are listed in **supplementary table 5** [↗](#).

4C Data Analysis

The sequenced reads in fastq file were demultiplexed by matching the appropriate primer sequences for each condition without allowing for any mismatches. Demultiplexed reads were processed using 4cseqpipe software. Restriction site tracks were created for the hg38 human genome by mentioning the restriction sites of the first cutter as GATC and that of the second cutter as CATG. The phred scores of demultiplexed reads were changed to phred64 format and the fastq files were converted into raw format. Further, valid 4C reads were mapped to the generated restriction site tracks. Unique fragment ends/non-unique fragment ends were used. The mapped reads were normalized and near cis domainograms at a maximum height of 0.1 were created by using the truncated log mean statistic with a trend resolution of 1kb for the genomic region chr11:42300000-42400000.

RNA Isolation, cDNA synthesis and PCR

Cells were lysed in 1 ml of Trizol (Thermo Fisher Inc.). 200 μ l chloroform was added to the sample, briefly vortexed and centrifuged at 12K rpm for 12 mins. The aqueous phase was carefully collected and transferred to the fresh tube. 1 volume of isopropanol was added to the sample and incubated at room temperature for 10 mins to precipitate the RNA. The samples were centrifuged at 12K rpm for 12 mins, supernatant was discarded without disturbing the pellet. The pellet obtained was washed with 75% ethanol. Pellet was air dried and dissolved in RNase free water. RNA obtained was treated with ezDNase (Invitrogen) to remove the traces of contaminating DNA. 1 μ g of RNA was used for each cDNA synthesis reaction by Superscript IV (Invitrogen) and random hexamers as per manufacturer's recommendation. The CFX96 touch (Biorad) real time PCR was used for qRT-PCRs. The fold changes were calculated by the $\Delta\Delta$ Ct method and individual expression data was normalized to GAPDH mRNA. The *p*-values were calculated by Student's unpaired two-tailed *t*-test from independent four biological replicates. qRT-PCR primers are listed in **Supplementary Table 6** [↗](#).

GRO-Seq analysis

Fastq files from GEO accession number GSE43836 were downloaded from European Nucleotide Archive. Reads with base quality <20 in a sliding window of 4 bases and with a length of <36 were removed using Trimmomatic 0.39. Trimmed reads were aligned using bowtie2 2.5.1 with default parameters. Duplicate reads from the alignment files were removed using samtools 1.16.1. De-duplicated aligned reads were assigned in a strand-specific manner to transcript feature of the hg38.ncbiRefSeq.gtf.gz file by allowing multi-mapping reads and considering the largest overlap in case of overlapping features using featureCounts v2.0.3. CPM normalised strand-specific signal files with a bin size of 1 bp were generated using bamCoverage 3.5.1. Differential gene expression analysis was performed with default parameters using Deseq2 1.36.0. Upregulated genes were defined as those with adjusted *p*-value <0.05 and log2(FC) >1 in their respective conditions. Top, Middle and bottom 10% of upregulated genes based on their base mean value in E2-40m vs VEH condition were subsetted as highly expressing, moderately expressing and low expressing upregulated genes. The closest upregulated genes in E2-160m vs E2-40m near highly expressing, moderately expressing and low expressing upregulated genes in E2-40m vs VEH condition were identified using bedtools closest v2.30.0. Only genes within a genomic distance of 1Mb were considered. Log2 transformed DEseq2 normalised counts with the addition of the arbitrary value 1 was used to compare the gene expression trends across time points and categories of upregulated genes. All plots were generated using R 4.2.2.

Acknowledgements

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Additional information

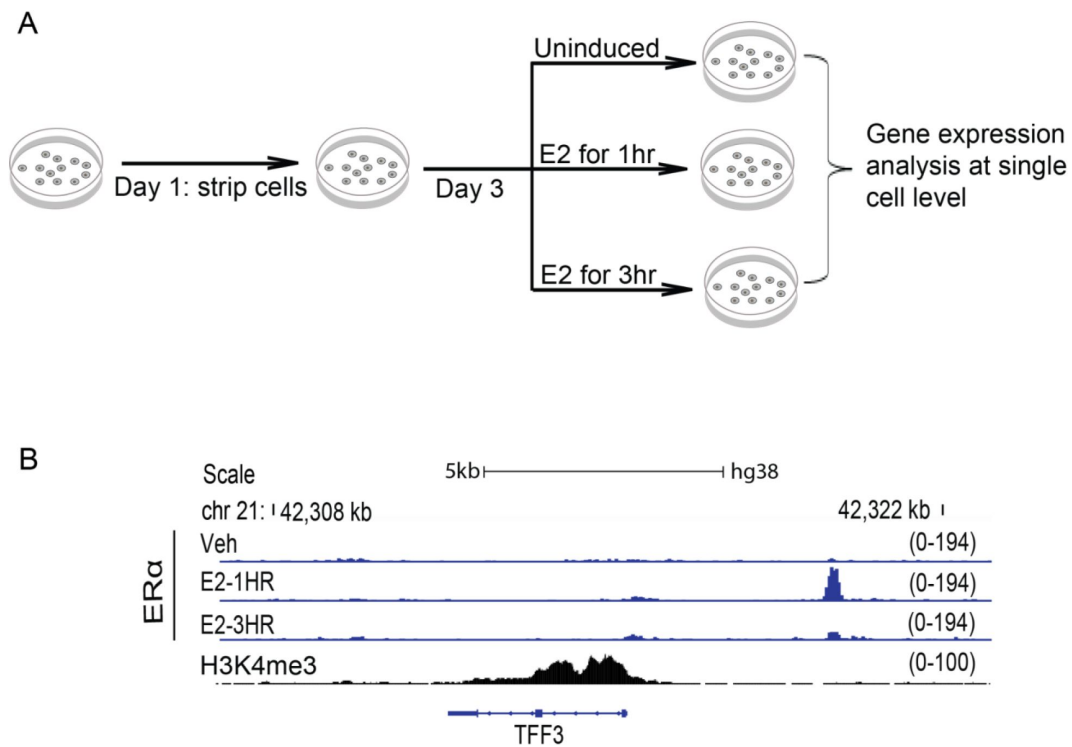
Contribution

The project was conceived by DN, AM, DB and ZI. Experiments were performed by DB and ZI. 4C and GRO-seq data analysis was performed by SN. The manuscript was written by DB, ZI, DN and AM. All authors approved the manuscript.

Competing Interest

Authors declare no competing interest.

Supplementary figure and tables

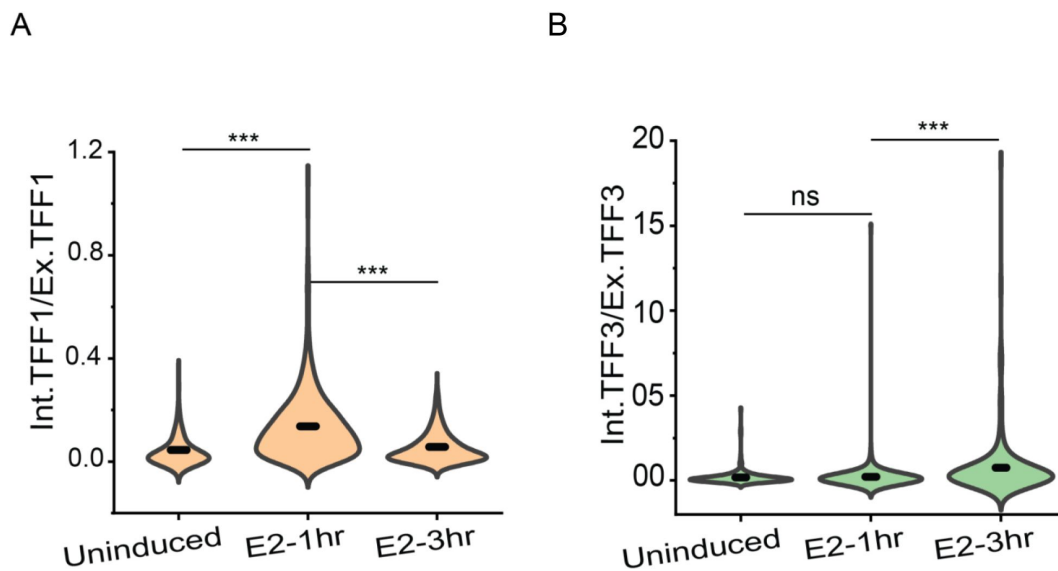


Supp. Figure 1.

Experimental design

A. Schematic depicting the experimental design. Cells were cultured in complete media for 24h, followed by stripping for 3 days. Finally, cells were induced for different durations with E2/Vehicle followed by different assays like smFISH, 4C-seq, smFISH-IF, etc.

B. Zoomed in the region around the *TFF3* gene, UCSC genome browser snapshots showing the binding of ERα and H3K4me3 signal. The first, second and third ERα ChIP-seq tracks are in vehicle-treated, E2 treated for 1h, and E2 treated for 3h in WT cells, respectively.

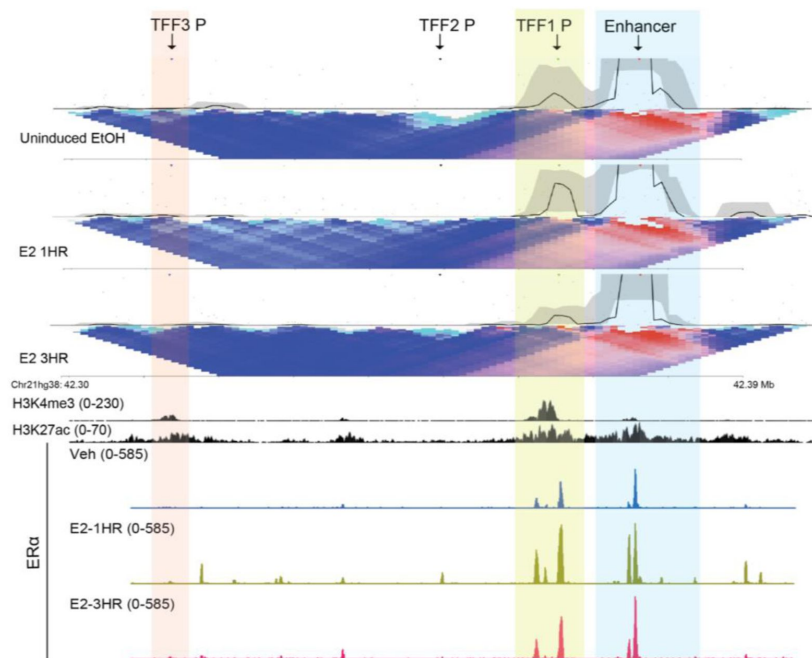


Supp. Figure 2.

***TFF1* and *TFF3* expressions show opposite trends during the E2-signaling time-course**

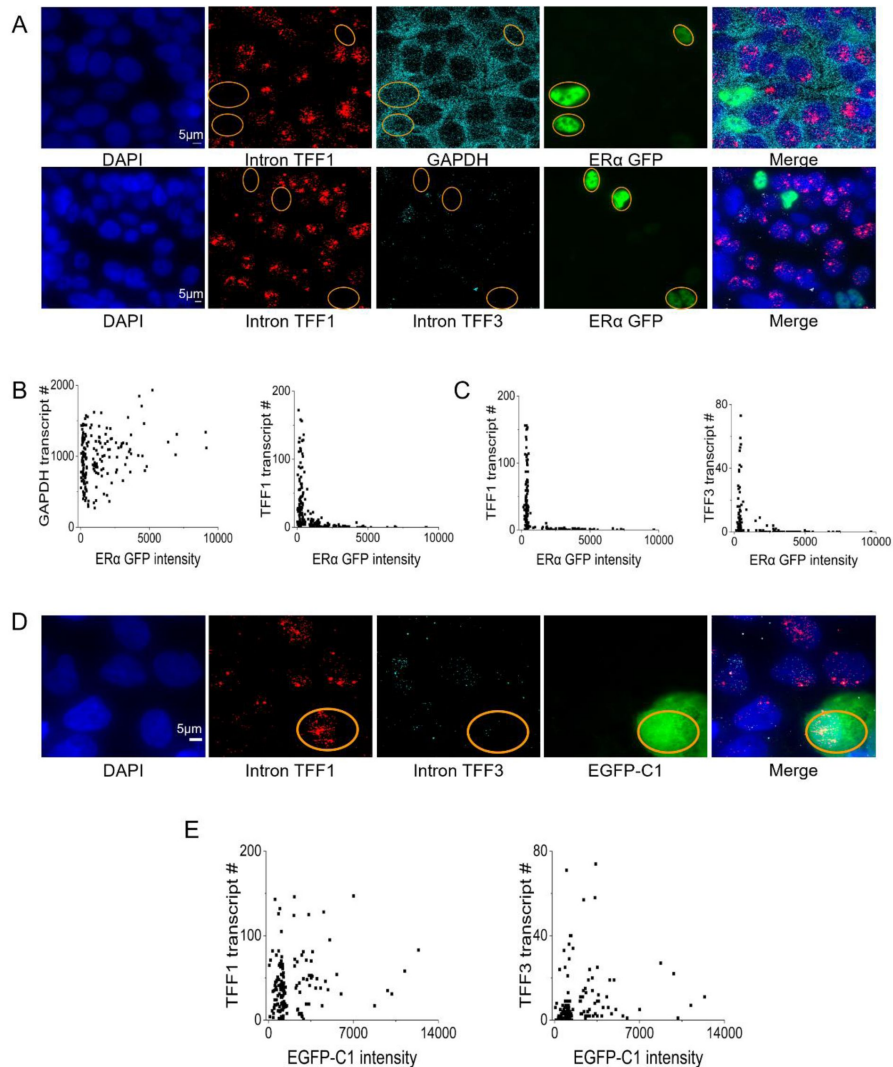
A. Violin plot showing the ratio of total intronic to absolute exonic *TFF1* counts are depicted. Absolute exonic counts are calculated by subtracting total intronic transcripts from total exonic transcripts. The graph shows the distribution of ratios combined from three different repeats. Error bars denote SEM. p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, and ns denotes $p > 0.05$.

B. Violin plot showing the ratio of total intronic to absolute exonic *TFF3* counts are depicted. Absolute exonic counts are calculated by subtracting total intronic transcripts from total exonic transcripts. The graph shows the distribution of ratios combined from three different repeats. Error bars denote SEM. p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, and ns denotes $p > 0.05$.



Supp. Figure 3.

TFF1 enhancer does not change target promoters during signalling time-course 4C-seq plot at *TFF1* enhancer viewpoint, the interaction with the promoter is highlighted in yellow. The plot is overlaid with H3K27ac, ERα ChIP signal, and gene annotations. There is no substantial interaction between the enhancer and *TFF3* locus at any time point, while the interaction between the enhancer and *TFF1* locus increases at 1hr and decreases at 3hr.



Supp. Figure 4.

Levels of ERα in the nucleus dictate the extent of *TFF1* and *TFF3* inductions

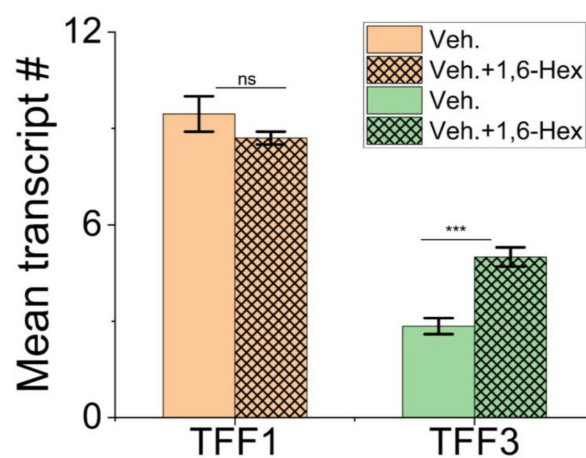
A. Representative images from smFISH experiments showing *InTFF1* and GAPDH (top panel) and *InTFF1* and *InTFF3* (bottom panel) in cells overexpressing ERα-GFP. Yellow circles show cells that are GFP positive, and the transcripts associated with them. Note the visibly fewer *TFF1*, and *TFF3* transcripts in the ERα-GFP positive cells, while GAPDH transcripts remain indistinguishable. This shows that ERα-GFP overexpression specifically affects the transcription of E2-regulated genes like *TFF1* and *TFF3* and not a housekeeping gene like GAPDH. This is further quantified in B and C.

B. Scatter plots showing the distribution of ERα-GFP intensities with GAPDH or *InTFF1* transcript counts on a cell-by-cell basis from the experiment in the first row of A. Cells with high ERα-GFP expression can have high GAPDH expression, but such cells necessarily have low *TFF1* counts.

C. Scatter plots showing the distribution of ERα GFP intensities with *InTFF1* or *InTFF3* counts on a cell-by-cell basis from the experiment in the second row of A. As in B, high ERα-GFP expression leads to low expression of E2-regulated genes like *TFF1* or *TFF3*.

D. Representative images showing smFISH for *InTFF1* and *InTFF3* in cells overexpressing EGFP-C1. Yellow circles show cells that are GFP positive, and the transcripts associated with them. Just expressing EGFP in the same backbone as the ERα-GFP plasmid, does not downregulate *TFF1*, and *TFF3* expression; indicating that the downregulation effect is specific to ERα overexpression and not a generic effect of cell transfection. This is quantified in E.

E. Scatter plots showing the distribution of EGFP-C1 intensities with *InTFF1* or *InTFF3* transcript counts on a cell-by-cell basis. There is no obvious correlation between the levels of EGFP-C1 and *TFF1* or *TFF3*, and a cell with high EGFP levels may well have large *TFF1* or *TFF3* transcript counts.



Supp. Figure 5.

LLPS perturbation down-regulates *TFF1* but supports *TFF3* expression

Mean transcript counts for *InTFF1* and *InTFF3* in control, and 3% 1,6-Hexanediol treated cells in the absence of E2-induction show no difference in *TFF1* counts on 1,6-Hexanediol treatment. The bar denotes the mean of means from two repeats, and the error bars denote SEM. p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

Probe Sequence (5' to 3')	Probe sequence name
catctttccttttcttgt	Int_TFF3_1
ctgaagtcagggttttctc	Int_TFF3_2
taaatgaaccttgctcctgc	Int_TFF3_3
ggcagggttgctatgcaatg	Int_TFF3_4
acttgggttaatggagcagc	Int_TFF3_5
ataaaagccatgccctcatt	Int_TFF3_6
tttgacctgctcagaagac	Int_TFF3_7
cgttaattgcttatcctctg	Int_TFF3_8
tcttacattcccttctctg	Int_TFF3_9
ggaaaatggcaatcacgtgc	Int_TFF3_10
cagaagttcactgcgttca	Int_TFF3_11
caagcctctatgtctgtatg	Int_TFF3_12
ccccaatatttcatgtttc	Int_TFF3_13
ctgaattgttgagcatgtcc	Int_TFF3_14
tgttttctactttttctgc	Int_TFF3_15
cgccatagagatcattgcaa	Int_TFF3_16
aaaccgtgacaggagaggca	Int_TFF3_17
cttgccttctgttttaaat	Int_TFF3_18
ccttcattgttgaccaaca	Int_TFF3_19
ccctgaaccagaaacaaa	Int_TFF3_20
cttctcgggaagtcacatac	Int_TFF3_21
catttgcctaagtgtcttcc	Int_TFF3_22
gtctgttaaatgctcagctc	Int_TFF3_23
ccttaggaagatgcactttc	Int_TFF3_24
cgagttcaaccactgctgaa	Int_TFF3_25
agtgtgttgcttcactttg	Int_TFF3_26
agcccagaaaggacaggagg	Int_TFF3_27
ttggaacaggtgtgtgtgtg	Int_TFF3_28
taagtgtgaagcctctgctc	Int_TFF3_29
aggggacagaagaggacagc	Int_TFF3_30
tgcgatgccatcatttgaa	Int_TFF3_31
gttactaaatcagttccctc	Int_TFF3_32
gtttccacttctctcaagaa	Int_TFF3_33
ggccgatttgcaaaaacagc	Int_TFF3_34
ggggatgccacaaatgacaa	Int_TFF3_35
cagagagaggagaggagat	Int_TFF3_36
cacgatgtggggcagctaag	Int_TFF3_37
agatggcttatagccaatcg	Int_TFF3_38
tgagctggcacttggacatg	Int_TFF3_39

Supp. table 1.

InTFF3 probe sequence

tggatgctggtccaggaag	Int_TFF3_40
cctaggacaaaggcaagag	Int_TFF3_41
catcattgatcaggagaccg	Int_TFF3_42
agtggagctgaaggaacagg	Int_TFF3_43
cactcagaggagagtgtagt	Int_TFF3_44
cagggatcaggcagataaca	Int_TFF3_45
tcgggttgctaagcaagata	Int_TFF3_46
atgggggaatagccaagatt	Int_TFF3_47
acagagcaaaggcttgtgtg	Int_TFF3_48

Supp. table 1. (continued)

Supp. table 2.

ExTFF3 probe sequence

Probe Sequence (5' to 3')	Probe sequence name
aaggggagaggtcagttca	Exn_TFF3_1
ccagacagttttctctcc	Exn_TFF3_2
ctgcatgcctttgtcaagc	Exn_TFF3_3
ttgtttgcttggggaaggc	Exn_TFF3_4
tgtttgcacagctgctctg	Exn_TFF3_5
aggaggcctcattatgca	Exn_TFF3_6
ctcaggactcgcttcattg	Exn_TFF3_7
cagcatgcagagcgctctg	Exn_TFF3_8
aggacagcaaggccaggac	Exn_TFF3_9
cacgtactcctcagcagag	Exn_TFF3_10
acggcacactggttgacag	Exn_TFF3_11
actccttgggggtgacatg	Exn_TFF3_12
cagggatcctggagtcaaa	Exn_TFF3_13
ggggcttgaaacaccaagg	Exn_TFF3_14
gtgcctcagaaggtgcatt	Exn_TFF3_15
caatcacagccgggcaagg	Exn_TFF3_16
gctgagatgaacagtgcct	Exn_TFF3_17
cgggagcaaaggacagaa	Exn_TFF3_18
actttcagcagaagcgctt	Exn_TFF3_19
gacatcaggctccagatat	Exn_TFF3_20
catgggacctttattcggt	Exn_TFF3_21
gaagaactgtcctcgggtg	Exn_TFF3_22
acctcagaaaagtctcaggc	Exn_TFF3_23
ccacgacgcagcagaaata	Exn_TFF3_24
cttagggagtctttctctg	Exn_TFF3_25
atccttgcatgcactgcag	Exn_TFF3_26
gtgccagtctggattcaaa	Exn_TFF3_27
ctaggctttcctgtgacgt	Exn_TFF3_28
gaagcggcacttacagtgt	Exn_TFF3_29
gtatttttctgcttccc	Exn_TFF3_30
ttattggcttgctgtcttc	Exn_TFF3_31
ggaccactttggaagacag	Exn_TFF3_32
gtagcgagagtggttgta	Exn_TFF3_33
accaggaatagtacaagt	Exn_TFF3_34
gcctgtttgtcataatgt	Exn_TFF3_35

Probe Sequence (5' to 3')	Probe sequence name
acagagcaggaagaagcacg	Int_TFF1_1
tgacactgggaggattgta	Int_TFF1_2
ccaacacttcctcttgaaa	Int_TFF1_3
aaagcactattctgagacc	Int_TFF1_4
gggagatgttgcatgaaca	Int_TFF1_5
catttggcctatctggatg	Int_TFF1_6
caggaacctgtaggagagg	Int_TFF1_7
atgagagaggtggcttgac	Int_TFF1_8
cagggaatatgggtgccag	Int_TFF1_9
caaccgatccacttgacaac	Int_TFF1_10
ttgagatgcaaacactccc	Int_TFF1_11
atccattctgcaggtaaagg	Int_TFF1_12
cacacagcatttcctgacag	Int_TFF1_13
aaagtgcagtcgcagatgc	Int_TFF1_14
aataagtattcagctcccc	Int_TFF1_15
ccacctgaaactgtacctta	Int_TFF1_16
aacgtagaagccctttctc	Int_TFF1_17
ctctggaacccttgctttg	Int_TFF1_18
ccacagcagagatcaagagg	Int_TFF1_19
agttcgttctgtacaccgag	Int_TFF1_20
aaagacgctctgagcctga	Int_TFF1_21
gtaacaggatcatggacctg	Int_TFF1_22
ttcacacacgaatgcacta	Int_TFF1_23
ctttgtctttgtcgcgatg	Int_TFF1_24
tcaggagggaagattccag	Int_TFF1_25
tggtgaggaggatcatcat	Int_TFF1_26
cactgtgggggtcaagagag	Int_TFF1_27
cgacgtgaaggatcatcgc	Int_TFF1_28
tgggaaggatccgtgttcag	Int_TFF1_29
gtctttggagaaagtgtcc	Int_TFF1_30

Supp. table 3.

InTFF1 probe sequence

cttgtctgcttgctatcag	Int_TFF1_31
ataggaggggaggaactcag	Int_TFF1_32
agcagggggtgaaaggag	Int_TFF1_33
taggatctggtggtgacag	Int_TFF1_34
cgcggaatcaaaggtctcag	Int_TFF1_35
tctaaccagagcttcagg	Int_TFF1_36
cgcaggatcttatgcaagag	Int_TFF1_37
agaggagggaagacagtga	Int_TFF1_38
aagcagggcagtgaggaaag	Int_TFF1_39
cagagggtacaagctgtgaa	Int_TFF1_40
attgcaggttaaggtgagagg	Int_TFF1_41
gtgtgttcaggaggtggaaa	Int_TFF1_42
ttcaccaggtgaagatggag	Int_TFF1_43
aggttgaacaaagcaggggg	Int_TFF1_44
atgtgcgagacaacgcaagg	Int_TFF1_45
tctgatggaggagaaaggca	Int_TFF1_46
ttgtggactttgcatcttc	Int_TFF1_47
gctcagggaatgcaaggt	Int_TFF1_48

Supp. table 3. (continued)

Supp. table 4.

ExTFF1 probe sequence

Probe Sequence (5' to 3')	Probe sequence name
gcgacccccgagtcagggat	Exn_TFF1_1
tgccctctctctgtccaa	Exn_TFF1_2
cttgcttccatgggtggcc	Exn_TFF1_3
caggaccaggggcgagatc	Exn_TFF1_4
gccgagggccagcatggac	Exn_TFF1_5
tgtctgggcctcgccagg	Exn_TFF1_6
ggggccactgtacagtct	Exn_TFF1_7
ccacaattctgtcttcac	Exn_TFF1_8
gagggcggtgacaccaggaa	Exn_TFF1_9
cagcccttatttgcacact	Exn_TFF1_10
cacgaacgggtgctcgaa	Exn_TFF1_11
gatagaagcaccaggggac	Exn_TFF1_12
gagggacgtcgatgtatt	Exn_TFF1_13
tcgtataaaaaggccatac	Exn_TFF1_14
attctaaatcttcagaacc	Exn_TFF1_15
gtcttaaatgacttttcta	Exn_TFF1_16
atgctgatcagagcctctg	Exn_TFF1_17
tgtgtaaaggcatagctgg	Exn_TFF1_18
caccactggcgccgtgac	Exn_TFF1_19
gactcaggctacccattg	Exn_TFF1_20
ttcctggacctgaatgcag	Exn_TFF1_21
tccttagccctgccttc	Exn_TFF1_22
gtctaaaattcacactct	Exn_TFF1_23
caggcagatccctgcagaa	Exn_TFF1_24
ggacggcaccgcgtcagga	Exn_TFF1_25
tgggactaatcacctgct	Exn_TFF1_26
ggtggagggtggcagccgag	Exn_TFF1_27
agaagcgtgtctgaggtgt	Exn_TFF1_28
tgtgagccgaggcacagct	Exn_TFF1_29
gtcagagcagtcaatctgt	Exn_TFF1_30
ggccaattttgagtagtca	Exn_TFF1_31
atcgatctcttttaattt	Exn_TFF1_32

Supp. table 5.

4C Seq oligo sequences

Sequence name	Sequence (5' to 3')
4C <i>TFF1</i> Enhancer barcode1 F	AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACACGACG CTCTTCCGATCTAGAAGTCCTTGAGTGGGGAGATGATC
4C <i>TFF1</i> Enhancer barcode2 F	AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACACGACG CTCTTCCGATCTAGCGACCCTTGAGTGGGGAGATGATC
4C <i>TFF1</i> Enhancer barcode3 F	AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACACGACG CTCTTCCGATCTCCAATTCCTTGAGTGGGGAGATGATC
4C <i>TFF1</i> R	CAAGCAGAAGACGGCATAAGGTCGCAGTGACTGGAGTTC AGACGTGTGCTCTTCCGATCTCCGCTGTTGCGACACACCAGAGAGC

Supp. table 6.

qPCR oligo sequences

Sequence name	Sequence (5' to 3')
<i>TFF1</i> US F	GCCCTTATTTGCACACTGGG
<i>TFF1</i> US R	CATCAAGGAATTCAGCCAC
<i>TFF3</i> US F	CAATCACAGCCGGGCAAGGGT
<i>TFF3</i> US R	GGGAGGAGGCAGCACTAGGT
GAPDH F	CGCTCTCTGCTCCTCCTGTT
GAPDH R	CCATGGTGTCTGAGCGATGT

References

- Ahmed J., Meszaros A., Lazar T., Tompa P (2021) **DNA-binding domain as the minimal region driving RNA-dependent liquid-liquid phase separation of androgen receptor** *Protein Science* **30**:1380–1392 <https://doi.org/10.1002/pro.4100>
- Boehning M. *et al.* (2018) **RNA polymerase II clustering through carboxy-terminal domain phase separation** *Nature Structural & Molecular Biology* **25**:833–840 <https://doi.org/10.1038/s41594-018-0112-y>
- Boija A. *et al.* (2018) **Transcription Factors Activate Genes through the Phase-Separation Capacity of Their Activation Domains** *Cell* **175**:1842–1855 <https://doi.org/10.1016/j.cell.2018.10.042>
- Bojszuk D., Nagy G., Balint B. L (2017) **Inducible super-enhancers are organized based on canonical signal-specific transcription factor binding elements** *Nucleic Acids Research* **45**:3693–3706 <https://doi.org/10.1093/nar/gkw1283>
- Brown A. M., Jeltsch J. M., Roberts M., Chambon P (1984) **Activation of pS2 gene transcription is a primary response to estrogen in the human breast cancer cell line MCF-7** *Proceedings of the National Academy of Sciences* **81**:6344–6348 <https://doi.org/10.1073/pnas.81.20.6344>
- Buecker C., Srinivasan R., Wu Z., Calo E., Acampora D., Faial T., Simeone A., Tan M., Swigut T., Wysocka J (2014) **Reorganization of enhancer patterns in transition from naive to primed pluripotency** *Cell Stem Cell* **14**:838–853 <https://doi.org/10.1016/j.stem.2014.04.003>
- Bulger M., Groudine M (1999) **Looping versus linking: Toward a model for long-distance gene activation** *Genes & Development* **13**:2465–2477
- Cai D., Feliciano D., Dong P., Flores E., Gruebele M., Porat-Shliom N., Sukenik S., Liu Z., Lippincott-Schwartz J (2019) **Phase separation of YAP reorganizes genome topology for long-term YAP target gene expression** *Nature Cell Biology* **21**:1578–1589 <https://doi.org/10.1038/s41556-019-0433-z>
- Chalbos D., Philips A., Galtier F., Rochefort H (1993) **Synthetic antiestrogens modulate induction of pS2 and cathepsin-D messenger ribonucleic acid by growth factors and adenosine 3',5'-monophosphate in MCF7 cells** *Endocrinology* **133**:571–576 <https://doi.org/10.1210/en.133.2.571>
- Chen H., Levo M., Barinov L., Fujioka M., Jaynes J. B., Gregor T (2018) **Dynamic interplay between enhancer-promoter topology and gene activity** *Nature Genetics* **50**:1296–1303 <https://doi.org/10.1038/s41588-018-0175-z>
- Chepelev I., Wei G., Wangsa D., Tang Q., Zhao K (2012) **Characterization of genome-wide enhancer-promoter interactions reveals co-expression of interacting genes and modes of higher order chromatin organization** *Cell Research* **22**:490–503 <https://doi.org/10.1038/cr.2012.15>

- Chinery R., Williamson J., Poulson R (1996) **The Gene Encoding Human Intestinal Trefoil Factor (TFF3) Is Located on Chromosome 21q22.3 Clustered with Other Members of the Trefoil Peptide Family** *Genomics* **32**:281–284 <https://doi.org/10.1006/geno.1996.0117>
- Cho W.-K., Spille J.-H., Hecht M., Lee C., Li C., Grube V., Cisse I. I (2018) **Mediator and RNA polymerase II clusters associate in transcription-dependent condensates** *Science* **361**:412–415 <https://doi.org/10.1126/science.aar4199>
- Chong S. *et al.* (2018) **Imaging dynamic and selective low-complexity domain interactions that control gene transcription** *Science* **361** <https://doi.org/10.1126/science.aar2555>
- Chong S., Graham T. G. W., Dugast-Darzacq C., Dailey G. M., Darzacq X., Tjian R (2022) **Tuning levels of low-complexity domain interactions to modulate endogenous oncogenic transcription** *Molecular Cell* **82**:2084–2097 <https://doi.org/10.1016/j.molcel.2022.04.007>
- Clark M. B. *et al.* (2015) **Quantitative gene profiling of long noncoding RNAs with targeted RNA sequencing** *Nature Methods* **12**:339–342 <https://doi.org/10.1038/nmeth.3321>
- Conesa A. *et al.* (2016) **A survey of best practices for RNA-seq data analysis** *Genome Biology* **17** <https://doi.org/10.1186/s13059-016-0881-8>
- Coté A. *et al.* (2023) **Post-transcriptional splicing can occur in a slow-moving zone around the gene** *eLife* **12** <https://doi.org/10.7554/eLife.91357.2>
- Coulon A., Ferguson M. L., de Turris V., Palangat M., Chow C. C., Larson D. R. (2014) **Kinetic competition during the transcription cycle results in stochastic RNA processing** *eLife* **3** <https://doi.org/10.7554/eLife.03939>
- Dixon B. S., Sharma R. V., Dennis M. J (1996) **The Bradykinin B2 Receptor Is a Delayed Early Response Gene for Platelet-derived Growth Factor in Arterial Smooth Muscle Cells *** *Journal of Biological Chemistry* **271**:13324–13332 <https://doi.org/10.1074/jbc.271.23.13324>
- Drexler H. L., Choquet K., Churchman L. S (2020) **Splicing Kinetics and Coordination Revealed by Direct Nascent RNA Sequencing through Nanopores** *Molecular Cell* **77**:985–998 <https://doi.org/10.1016/j.molcel.2019.11.017>
- Femino A. M., Fay F. S., Fogarty K., Singer R. H (1998) **Visualization of single RNA transcripts in situ** *Science (New York, N.Y.)* **280**:585–590 <https://doi.org/10.1126/science.280.5363.585>
- Fowler T., Sen R., Roy A. L (2011) **Regulation of Primary Response Genes** *Molecular Cell* **44**:348–360 <https://doi.org/10.1016/j.molcel.2011.09.014>
- Freter R. R., Alberta J. A., Hwang G. Y., Wrenthmore A. L., Stiles C. D (1996) **Platelet-derived Growth Factor Induction of the Immediate-early gene MCP-1 Is Mediated by NF-κB and a 90-kDa Phosphoprotein Coactivator *** *Journal of Biological Chemistry* **271**:17417–17424 <https://doi.org/10.1074/jbc.271.29.17417>
- Friedman M. J., Wagner T., Lee H., Rosenfeld M. G., Oh S (2024) **Enhancer-promoter specificity in gene transcription: Molecular mechanisms and disease associations** *Experimental & Molecular Medicine* **56**:772–787 <https://doi.org/10.1038/s12276-024-01233-y>
- Fullwood M. J. *et al.* (2009) **An oestrogen-receptor-α-bound human chromatin interactome** *Nature* **462**:58–64 <https://doi.org/10.1038/nature08497>

- Furlong E. E. M., Levine M (2018) **Developmental enhancers and chromosome topology** *Science* **361**:1341–1345 <https://doi.org/10.1126/science.aau0320>
- Galouzis C. C., Furlong E. E. M (2022) **Regulating specificity in enhancer-promoter communication** *Current Opinion in Cell Biology* **75** <https://doi.org/10.1016/j.ceb.2022.01.010>
- Gamliel A., Meluzzi D., Oh S., Jiang N., Destici E., Rosenfeld M. G., Nair S. J (2022) **Long-distance association of topological boundaries through nuclear condensates** *Proceedings of the National Academy of Sciences of the United States of America* **119** <https://doi.org/10.1073/pnas.2206216119>
- Hah N., Danko C. G., Core L., Waterfall J. J., Siepel A., Lis J. T., Kraus W. L (2011) **A rapid, extensive, and transient transcriptional response to estrogen signaling in breast cancer cells** *Cell* **145**:622–634 <https://doi.org/10.1016/j.cell.2011.03.042>
- Hah N., Murakami S., Nagari A., Danko C. G., Kraus W. L (2013) **Enhancer transcripts mark active estrogen receptor binding sites** *Genome Research* **23**:1210–1223 <https://doi.org/10.1101/gr.152306.112>
- Haimovich G., Gerst J. E (2018) **Single-molecule Fluorescence in situ Hybridization (smFISH) for RNA Detection in Adherent Animal Cells** *Bio-Protocol* **8** <https://doi.org/10.21769/BioProtoc.3070>
- Herschman H. R (1991) **Primary response genes induced by growth factors and tumor promoters** *Annual Review of Biochemistry* **60**:281–319 <https://doi.org/10.1146/annurev.bi.60.070191.001433>
- Jeltsch J. M., Roberts M., Schatz C., Garnier J. M., Brown A. M., Chambon P (1987) **Structure of the human oestrogen-responsive gene pS2** *Nucleic Acids Research* **15**:1401–1414
- Khodor Y. L., Menet J. S., Tolan M., Rosbash M (2012) **Cotranscriptional splicing efficiency differs dramatically between Drosophila and mouse** *RNA* **18**:2174–2186 <https://doi.org/10.1261/rna.034090.112>
- Kim J., Petz L. N., Ziegler Y. S., Wood J. R., Potthoff S. J., Nardulli A. M (2000) **Regulation of the estrogen-responsive pS2 gene in MCF-7 human breast cancer cells** *The Journal of Steroid Biochemistry and Molecular Biology* **74**:157–168 [https://doi.org/10.1016/S0960-0760\(00\)00119-9](https://doi.org/10.1016/S0960-0760(00)00119-9)
- Koşar Z., Erbaş A (2022) **Can the Concentration of a Transcription Factor Affect Gene Expression?** *Frontiers in Soft Matter* **2**
- Kroschwald S., Maharana S., Simon A (2017) **Hexanediol: A chemical probe to investigate the material properties of membrane-less compartments** *Matters* <https://doi.org/10.19185/matters.201702000010>
- Kwon S (2013) **Single-molecule fluorescence in situ hybridization: Quantitative imaging of single RNA molecules** *BMB Reports* **46**:65–72 <https://doi.org/10.5483/BMBRep.2013.46.2.016>
- Levenson A. S., Jordan V. C (1997) **MCF-7: The First Hormone-responsive Breast Cancer Cell Line** *Cancer Research* **57**:3071–3078
- Levine M., Tjian R (2003) **Transcription regulation and animal diversity** *Nature* **424**:147–151 <https://doi.org/10.1038/nature01763>

- Li W. *et al.* (2013) **Functional roles of enhancer RNAs for oestrogen-dependent transcriptional activation** *Nature* **498**:516–520 <https://doi.org/10.1038/nature12210>
- Liu Z. *et al.* (2014) **Enhancer Activation Requires trans-Recruitment of a Mega Transcription Factor Complex** *Cell* **159**:358–373 <https://doi.org/10.1016/j.cell.2014.08.027>
- Mann R., Notani D (2023) **Transcription factor condensates and signaling driven transcription** *Nucleus* **14** <https://doi.org/10.1080/19491034.2023.2205758>
- Masiakowski P., Breathnach R., Bloch J., Gannon F., Krust A., Chambon P (1982) **Cloning of cDNA sequences of hormone-regulated genes from the MCF-7 human breast cancer cell line** *Nucleic Acids Research* **10**:7895–7903 <https://doi.org/10.1093/nar/10.24.7895>
- Nair S. J. *et al.* (2019) **Phase separation of ligand-activated enhancers licenses cooperative chromosomal enhancer assembly** *Nature Structural & Molecular Biology* **26**:193–203 <https://doi.org/10.1038/s41594-019-0190-5>
- Nunez A. M., Berry M., Imler J. L., Chambon P (1989) **The 5' flanking region of the pS2 gene contains a complex enhancer region responsive to oestrogens, epidermal growth factor, a tumour promoter (TPA), the c-Ha-ras oncoprotein and the c-jun protein** *The EMBO Journal* **8**:823–829
- Oh S. *et al.* (2021) **Enhancer release and retargeting activates disease-susceptibility genes** *Nature* **595**:735–740 <https://doi.org/10.1038/s41586-021-03577-1>
- Panigrahi A., O'Malley B. W (2021) **Mechanisms of enhancer action: The known and the unknown** *Genome Biology* **22** <https://doi.org/10.1186/s13059-021-02322-1>
- Ptashne M (1986) **Gene regulation by proteins acting nearby and at a distance** *Nature* **322**:697–701 <https://doi.org/10.1038/322697a0>
- Quintin J. *et al.* (2014) **Dynamic Estrogen Receptor Interactomes Control Estrogen-Responsive Trefoil Factor (TFF) Locus Cell-Specific Activities** *Molecular and Cellular Biology* **34**:2418–2436 <https://doi.org/10.1128/MCB.00918-13>
- Raj A., Peskin C. S., Tranchina D., Vargas D. Y., Tyagi S (2006) **Stochastic mRNA Synthesis in Mammalian Cells** *PLOS Biology* **4** <https://doi.org/10.1371/journal.pbio.0040309>
- Raj A., van den Bogaard P., Rifkin S. A., van Oudenaarden A., Tyagi S. (2008) **Imaging individual mRNA molecules using multiple singly labeled probes** *Nature Methods* **5**:877–879 <https://doi.org/10.1038/nmeth.1253>
- Rao S.S.P., *et al.* (2014) **A 3D map of the human genome at kilobase resolution reveals principles of chromatin looping** *Cell* **159**:1665–1680 <https://doi.org/10.1016/j.cell.2014.11.021>
- Richter W. F., Nayak S., Iwasa J., Taatjes D. J (2022) **The Mediator complex as a master regulator of transcription by RNA polymerase II** *Nature Reviews Molecular Cell Biology* **23**:732–749 <https://doi.org/10.1038/s41580-022-00498-3>
- Ryu K., Park G., Cho W.-K (2024) **Emerging insights into transcriptional condensates** *Experimental & Molecular Medicine* **56**:820–826 <https://doi.org/10.1038/s12276-024-01228-9>
- Sabari B. R. *et al.* (2018) **Coactivator condensation at super-enhancers links phase separation and gene control** *Science* **361** <https://doi.org/10.1126/science.aar3958>

Sanyal A., Lajoie B. R., Jain G., Dekker J (2012) **The long-range interaction landscape of gene promoters** *Nature* **489**:109–113 <https://doi.org/10.1038/nature11279>

Saravanan B. *et al.* (2020) **Ligand dependent gene regulation by transient ERα clustered enhancers** *PLOS Genetics* **16** <https://doi.org/10.1371/journal.pgen.1008516>

Sha Y., Phan J. H., Wang M. D (2015) **Effect of low-expression gene filtering on detection of differentially expressed genes in RNA-seq data** *Conference Proceedings : ... Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual Conference* **2015** <https://doi.org/10.1109/EMBC.2015.7319872>

Shi Y., Chen J., Zeng W.-J., Li M., Zhao W., Zhang X.-D., Yao J (2021) **Formation of nuclear condensates by the Mediator complex subunit Med15 in mammalian cells** *BMC Biology* **19** <https://doi.org/10.1186/s12915-021-01178-y>

Shrinivas K. *et al.* (2019) **Enhancer Features that Drive Formation of Transcriptional Condensates** *Molecular Cell* **75**:549–561 <https://doi.org/10.1016/j.molcel.2019.07.009>

Skinner S. O., Xu H., Nagarkar-Jaiswal S., Freire P. R., Zwaka T. P., Golding I (2016) **Single-cell analysis of transcription kinetics across the cell cycle** *eLife* **5** <https://doi.org/10.7554/eLife.12175>

Stortz M., Pecci A., Presman D. M., Levi V (2020) **Unraveling the molecular interactions involved in phase separation of glucocorticoid receptor** *BMC Biology* **18** <https://doi.org/10.1186/s12915-020-00788-2>

Stortz M., Presman D. M., Levi V (2024) **Transcriptional condensates: A blessing or a curse for gene regulation?** *Communications Biology* **7**:1–10 <https://doi.org/10.1038/s42003-024-05892-5>

Sun J.-M., Spencer V. A., Li L., Yu Chen H., Yu J., Davie J. R (2005) **Estrogen regulation of trefoil factor 1 expression by estrogen receptor α and Sp proteins** *Experimental Cell Research* **302**:96–107 <https://doi.org/10.1016/j.yexcr.2004.08.015>

Svensson V., Natarajan K. N., Ly L.-H., Miragaia R. J., Labalette C., Macaulay I. C., Cvejic A., Teichmann S. A (2017) **Power analysis of single-cell RNA-sequencing experiments** *Nature Methods* **14**:381–387 <https://doi.org/10.1038/nmeth.4220>

Tarazona S., García-Alcalde F., Dopazo J., Ferrer A., Conesa A (2011) **Differential expression in RNA-seq: A matter of depth** *Genome Research* **21**:2213–2223 <https://doi.org/10.1101/gr.124321.111>

Uyehara C. M., Apostolou E (2023) **3D enhancer-promoter interactions and multi-connected hubs: Organizational principles and functional roles** *Cell Reports* **42** <https://doi.org/10.1016/j.celrep.2023.112068>

van de Werken HJ, de Vree PJ, Splinter E, Holwerda SJ, Klous P, de Wit E, de Laat W (2012) **4C technology: protocols and data analysis** *Methods Enzymol* **513**:89–112 <https://doi.org/10.1016/B978-0-12-391938-0.00004-5>

Williams M., Lyu M.-S., Yang Y.-L., Lin E. P., Dunbrack R., Birren B., Cunningham J., Hunter K (1999) **Ier5, a Novel Member of the Slow-Kinetics Immediate-Early Genes** *Genomics* **55**:327–334 <https://doi.org/10.1006/geno.1998.5679>

- Winkles J. A., Moldave K. (1997) **Serum-and Polypeptide Growth Factor-Inducible Gene Expression in Mouse Fibroblasts** *Progress in Nucleic Acid Research and Molecular Biology* Academic Press :41–78 [https://doi.org/10.1016/S0079-6603\(08\)60033-1](https://doi.org/10.1016/S0079-6603(08)60033-1)
- Winston J. T., Pledger W. J (1993) **Growth factor regulation of cyclin D1 mRNA expression through protein synthesis-dependent and -independent mechanisms** *Molecular Biology of the Cell* **4**:1133–1144 <https://doi.org/10.1091/mbc.4.11.1133>
- Yamamoto K. R., Alberts B. M (1976) **Steroid Receptors: Elements for Modulation of Eukaryotic Transcription** *Annual Review of Biochemistry* **45**:721–746 <https://doi.org/10.1146/annurev.bi.45.070176.003445>
- Yan J. *et al.* (2018) **Histone H3 lysine 4 monomethylation modulates long-range chromatin interactions at enhancers** *Cell Research* **28**:204–220 <https://doi.org/10.1038/cr.2018.1>
- Zabidi M. A., Arnold C. D., Schernhuber K., Pagani M., Rath M., Frank O., Stark A (2015) **Enhancer–core-promoter specificity separates developmental and housekeeping gene regulation** *Nature* **518**:556–559 <https://doi.org/10.1038/nature13994>
- Zenkhusen D., Larson D. R., Singer R. H (2008) **Single-RNA counting reveals alternative modes of gene expression in yeast** *Nature Structural & Molecular Biology* **15**:1263–1271 <https://doi.org/10.1038/nsmb.1514>
- Zhang F. *et al.* (2023) **Dynamic phase separation of the androgen receptor and its coactivators key to regulate gene expression** *Nucleic Acids Research* **51**:99–116 <https://doi.org/10.1093/nar/gkac1158>
- Zhou T., Feng Q. (2022) **Androgen receptor signaling and spatial chromatin organization in castration-resistant prostate cancer** *Frontiers in Medicine* **9**

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Reviewer #1 (Public review):

Summary:

The manuscript by Bohra et al. describes the indirect effects of ligand-dependent gene activation on neighboring non-target genes. The authors utilized single-molecule RNA-FISH (targeting both mature and intronic regions), 4C-seq, and enhancer deletions to demonstrate that the non-enhancer-targeted gene TFF3, located in the same TAD as the target gene TFF1, alters its expression when TFF1 expression declines at the end of the estrogen signaling peak. Since the enhancer does not loop with TFF3, the authors conclude that mechanisms other than estrogen receptor or enhancer-driven induction are responsible for TFF3 expression. Moreover, ER α intensity correlations show that both high and low levels of ER α are unfavorable for TFF1 expression. The ER α level correlations are further supported by overexpression of GFP-ER α . The authors conclude that transcriptional machinery used by TFF1 for its acute activation can negatively impact the TFF3 at peak of signaling but once, the condensate dissolves, TFF3 benefits from it for its low expression.

Strengths:

The findings are indeed intriguing. The authors have maintained appropriate experimental controls, and their conclusions are well-supported by the data.

Weaknesses:

There are some major and minor concerns that related to approach, data presentation and discussion. But I think they can be fixed with more efforts.

<https://doi.org/10.7554/eLife.102417.1.sa1>

Reviewer #2 (Public review):**Summary:**

In this manuscript by Bohra et al., the authors use the well-established estrogen response in MCF7 cells to interrogate the role of genome architecture, enhancers, and estrogen receptor concentration in transcriptional regulation. They propose there is competition between the genes TFF1 and TFF3 which is mediated by transcriptional condensates. This reviewer does not find these claims persuasive as presented. Moreover, the results are not placed in the context of current knowledge.

Strengths:

High level of ERalpha expression seems to diminish the transcriptional response. Thus, the results in Fig. 4 have potential insight into ER-mediated transcription. Yet, this observation is not pursued in great depth however, for example with mutagenesis of ERalpha. However, this phenomenon - which falls under the general description of non monotonic dose response - is treated at great depth in the literature (i.e. PMID: 22419778). For example, the result the authors describe in Fig. 4 has been reported and in fact mathematically modeled in PMID 23134774. One possible avenue for improving this paper would be to dig into this result at the single-cell level using deletion mutants of ERalpha or by perturbing co-activators.

Weaknesses:

There are concerns with the smRNA FISH experiments. It is highly unusual to see so much intronic signal away from the site of transcription (Fig. 2) (PMID: 27932455, 30554876) which suggests to me the authors are carrying out incorrect thresholding or have a substantial amount of labeling background. The Cote paper cited in the manuscript is likewise inconsistent with their findings and is cited in a misleading manner: they see splicing within a very small region away from the site of transcription.

One substantial way to improve the manuscript is to take a careful look at previous single cell analysis of the estrogen response, which in some cases has been done on the exact same genes (PMID: 29476006, 35081348, 30554876, 31930333). In some of these cases, the authors reach different conclusions than those presented in the present manuscript. Likewise, there have been more than a few studies which characterized these enhancers (the first one I know of is: PMID 18728018). Also, Oh et al. 2021 (cited in the manuscript) did show an interaction between TFF1e and TFF3, which seems to contradict the conclusion from Fig. 3. In summary, the results of this paper are not in dialog with the field, which is a major shortcoming.

In the opinion of this reviewer, there are few - if any - experiments to interrogate the existence of LLPS for diffraction limited spots such as those associated with transcription. This difficulty is a general problem with the field and not specific to the present manuscript. For example, transient binding will also appear as a dynamic 'spot' in the nucleus, independently of any higher order interactions. As for Fig. 5, I don't think treating cells with 1,6 hexanediol is any longer considered a credible experiment. For example, there are profound effects on chromatin independent of changes in LLPS (PMID: 33536240).

Summary:

In conclusion, I suggest that the authors look at alternative explanations and analyses -- many of which are experimentally and mathematically rigorous and pre-date the condensate model -- to explain their data.

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